

Module 5: Heredity

Reproduction

IQ 1: How does reproduction ensure the continuity of a species?

- explain the mechanisms of reproduction that ensure the continuity of a species, by analysing sexual and asexual methods of reproduction in a variety of organisms, including but not limited to:
 - animals: advantages of external and internal fertilisation
 - plants: asexual and sexual reproduction
 - fungi: budding, spores
 - bacteria: binary fission
 - protists: binary fission, budding
- analyse the features of fertilisation, implantation and hormonal control of pregnancy and birth in mammals
- evaluate the impact of scientific knowledge on the manipulation of plant and animal reproduction in agriculture

Sexual VS Asexual Reproduction

- Sexual Reproduction
 - Requires 1-2 parents and involves union of gametes
 - Requires meiosis and fertilisation
 - Offsprings are genetically different to the parents
- Asexual Reproduction
 - 1 parent
 - Genetically identical offspring to parents
 - No meiosis, no gametes, no fertilisation

	Asexual	Sexual
Advantages	• Does not require a lot of energy • Offspring produced quickly • Creates large, thriving populations in stable (often harsh) environments where organisms have to be so specific to environment that there is often little benefit to GV	• Genetic information from both parents so introduces genetic variation • Able to live in variety of environmental settings • Able to adapt to changes in the environment
Disadvantages	• Limited ability to adapt • Face mass die-offs if environment changes – particularly vulnerable to drought, disease or new predators → only few survive if this occurs	• Requires increased time and energy • Produce small populations

Internal VS External Fertilization

External:

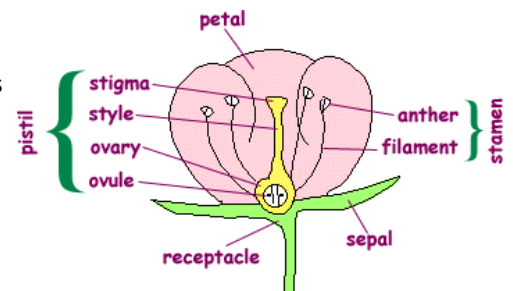
Advantages	Disadvantage
• Results in increased genetic variation • Produces an increased number of offspring • Gametes released can drift; therefore easier to find a mate • Wide dispersal of offspring	• Large amounts of gametes are wasted • Chances of fertilisation decreased due to predators and environmental hazards • Eggs and sperms may not come in contact • Desiccation of zygote/gametes

Internal:

Advantages	Disadvantage
• Increased chance of gamete fusion • Increased successful fertilization • Mates are selective • Decreased desiccation • Fewer eggs required for survival of sufficient number of offspring • Young protected against predators/environment • Developing young are fed and protected, therefore greater survival chances	• Fewer offsprings (fewer progeny) • Requires increased energy/time, therefore greater parental care • Resource competition for mates • More complex mating rituals • Potential for spread of sexually transmitted infections throughout population

Reproduction in the Plant Kingdom

- All plants reproduce sexually, some with asexual methods as well
- Relies on successful fusion of male and female gametes and thus rely on external agents to carry gametes from one parent to another (pollinating agents)
- Parents also rely on external agents to disperse their seeds
- 2 commonly observed structures for sexual reproduction
 - Gymnosperms = seed cones
 - Angiosperms = flowers



Plant Anatomy

Stamen (male)

- Anther – where pollen grains are formed
- Filament – stalk that carries the anther

Pistil (female)

Made of 1 or more carpels fused together

- Stigma – sticky top surface to which pollen adheres to
- Style – joins the stigma to the ovary
- Ovary – where ovules are formed

Pollination

- External agents carry pollen (eg. birds, bees, wind, water, etc.)
- Pollination (transfer of male gamete (pollen)) to female gamete
- Pollen released from anther and transferred to stigma
- Travels down style to the ovary and then to ovule
- Fertilisation – joining of gametes → embryo → seed → Seed dispersed
- Germinates → develops into new plant

Self-pollination VS Cross-pollination

- Self-pollination: requires less energy as no need for plant to produce structures to attract pollinators (eg. brightly colour petals)
- Cross-pollination rely on external agents to transfer pollen from anthers to stigmas of another plant (more GV)

Forms of Asexual Reproduction

Plants: Vegetative Propagation

- Some adult plants produce vegetative organs such as runners, bulbs, tubers, rhizomes and suckers, from which new plants can arise → equivalent to cloning
- These organs allow plants to survive in adverse conditions, eg. cold weather, where, even if the plant above ground dies, these organs continue growing
- Used in agriculture to increase production of crops when seeds are unavailable/difficult to germinate (eg. watermelons, mangoes)
- Also used if plants have specific desirable traits that farmers want to perpetuate in future generations

Fungi: Budding and Spores

- Genetically identical smaller buds begin as outgrowths from the parent plant, which separate from parent and grow into a new individual, eg. yeast
- Leaves a bud scar on parent cell and birth scar on daughter cell from cytokinesis
- Division is unequal, buds smaller than parents
- Reproduce via spores (unicellular reproductive cells)
 - *Mitospores, asexual*: haploid and genetically identical to parent, produced by mitosis
 - Produced and contained in a structure called a sporangium, from which spores are released into the environment
 - *Meiospores, sexual*: diploid and genetically different to parents, produced by meiosis
 - Leads to genetic variation in a fungal species

Prokaryotes: Binary Fission

- Bacteria and archaea have no nuclei. DNA is a singular, circular chromosome w/ origin of replication
- A newly divided cell grows to twice its size, replicates its DNA from origin, and splits into two cells with identical genetic material
- Relatively quick (bacteria can double in number in 20 minutes in favourable environments)
- Daughter cells approximately equal sizes
- Daughter cells do not have birth/bud scars like budding
- Binary fission is very efficient. Populations multiply quickly and exponentially, but are limited by nutrient availability. Populations typically cycle through lag, growth, stationary, and death phases, depending on nutrient availability.

Protists: Binary Fission and Budding

- Mainly reproduce by asexually binary fission, involving mitosis and the formation of a spindle within the cytoplasm of the cell to distribute chromosomes equally
- Varies from hours to days
- Transverse (split vertically) or longitudinal (split horizontally)
- Multiple fissions also occur in the same species
- Can also reproduce by budding

Asexual Reproduction in Animals

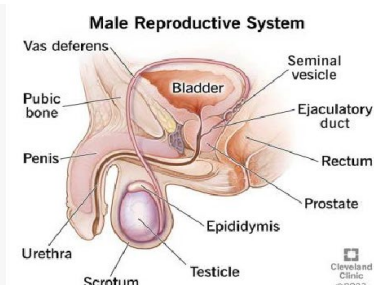
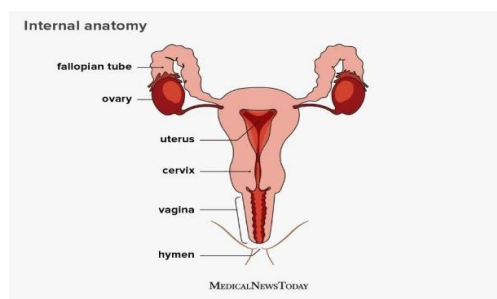
- Advantageous when mates are hard to find
- Some can alternate between sexual and asexual, gaining advantages from both and avoiding disadvantages

Types of Mammals

- Sexual reproduction produces genetically different offspring to ensure continuity of species
- Viviparous – give birth to live young
- Oviparous – lay eggs in which their young develop
- Haploid (n) – 1/2 chromosomes of a full cell
- Diploid (2n) – full set of chromosomes for the species

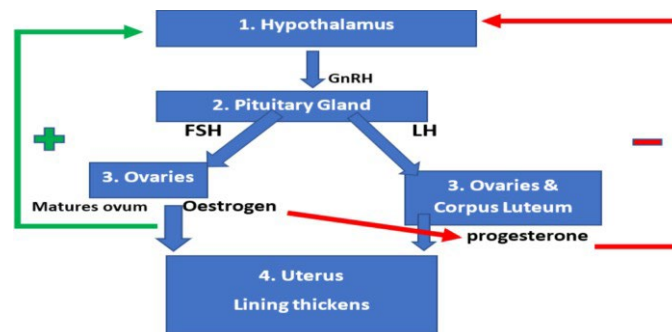
Placental	Marsupial	Monotreme
Uterus provides nutrients and protection	Undeveloped joey is protected and nourished in external pouch after an early birth, allowing another fertilisation to occur	Lay leathery eggs and the baby develops inside

Reproductive Systems



Hormonal Control of Female Reproductive Cycle

- Hormones – chemical substances that act as messengers in coordinating bodily function



- Oestrogen and progesterone levels regulate ovarian and menstrual cycles and create ideal conditions for ovulation, implantation and maintaining pregnancy
- Oxytocin is released to stimulate uterine contractions and birth

Ovulation – Production of egg

- FSH: Follicle Stimulating Hormone
 - Causes 1+ oocytes (immature egg cells) to undergo meiosis and mature in a follicle in the ovary
- Follicles
 - Follicles containing the maturing egg release oestrogen, which causes the endometrium to become thicker, softer, spongy and rich in blood vessels in preparation to receive a fertilised egg
 - Oestrogen acts on the pituitary gland
- Ovulation
 - Release of a mature egg, triggered by the release of luteinising hormone (LH) which is received by the pituitary gland in the brain
 - The ovum bursts out of the follicle and travels via the fallopian tubes towards uterus (burst of follicle = corpus luteum)
 - Stimulated by LH and secretes large amounts of oestrogen and progesterone which cause further thickening of uterus lining

Internal Fertilisation

- Sperm attracted to egg by rheotaxis (movement through a fluid), with the oviducts secreting a fluid that travels down the female reproductive tract that sperm swims upstream in
 - Presence of progesterone and an alkaline pH causes sperm to mature so they can penetrate the egg, making them hypermobile and causing their tails to beat strongly to propel
- Sperm use enzymes (both from the follicle cells attached to the egg and released from the sperm) to dissolve and physically penetrate the outer layers of the egg to reach the cell membrane
 - Surface proteins only allow one sperm to penetrate the cell membrane
 - Molecules on the sperm surface bind to receptors on the egg's cell membrane (to ensure that sperm from the same species fertilise the egg) then the nucleus of the sperm enters the cytoplasm of the egg cell
 - Fusion of haploid egg/sperm nuclei results in a diploid zygote cell (fertilised egg) as the first sperm to penetrate the inner barrier causes the ovum to immediately undergo its second meiotic division

Implantation

- Zygote travels from oviduct to uterus
 - Cleavage occurs (rapid cell proliferation) and the single celled zygote divides into 100's of smaller cells by mitosis
- 3-4 days later, the morula has formed (16 cells), enters uterus
 - Unspecialised ball of embryonic stem cells
- Blastocyst: develops from morula
 - Cells begin to differentiate
 - Attaches to uterine wall by day 8/9
- Gastrula forms from blastocyst over approx. 5 days and has 3 different layers of cells
 - Eventually becomes an embryo and then foetus with all basic adult features
- Placenta is formed from outer layer of cells of blastocyst and by 5th week, umbilical cord develops from remnants of egg's yolk sac
 - Becomes source of nutrients, remains attached until after birth
- Placenta is an exchange organ
 - Brings blood vessels of foetus close to maternal blood
 - Nutrients from mum's blood diffuses across into umbilical vein and into foetus
 - Waste from foetus diffuses other direction
 - Source of hormones to maintain the pregnancy

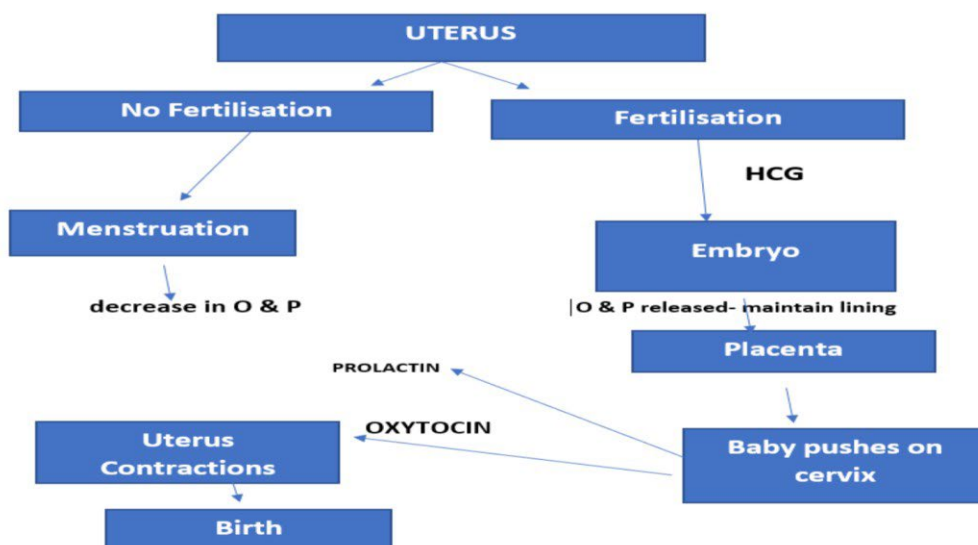
Development of the Embryo

- Major organs form from 3 main layers of the gastrula, completed by week 8, known as foetus
- 3 main layers:
 - Ectoderm: outermost, forms epidermis, hair, peripheral NS, brain, spinal cord cells
 - Mesoderm: middle, forms muscle, cartilage, kidney, gonad cells
 - Endoderm: innermost, forms lungs, bladder, lining of digestive system (stomach, colon, liver, pancreas)

Development of Foetus

- From 8-38 weeks, age calculated from date of mum's last menstrual period
- Foetus grows and organs continue to develop
- Foetus protected by amniotic cavity

Hormonal Control of Pregnancy and Birth



Hormone	Secreted by	Active period during pregnancy	Effect
Oestrogen	Corpus Luteum, mature follicle, placenta	Whole pregnancy	Thicken uterine lining (endometrium) & trigger production of GnRH
Progesterone	Corpus Luteum, placenta	Whole pregnancy	Stabilise uterine lining (endometrium) & suppress immunity to baby's antigens <i>Note: combined oestrogen & progesterone inhibits GnRH production</i>
Human Chorionic Gonadotropin (HCG)	Developing Embryo	1 st trimester	Maintain corpus luteus to secrete oestrogen & progesterone, which help maintain the endometrium
Oxytocin	Pituitary (of mother and baby)	Labour	Softens cervix ready for birth & causes contractions of uterus
Prolactin	Pituitary	2 nd trimester	Milk production & mammary
Follicle Stimulating Hormone (FSH)	Pituitary	Ovulation	Stimulates follicles to grow in ovaries
Luteinising Hormone (LH)	Pituitary	Ovulation	Cause egg to burst out of mature follicle (ovulation) & cause remnants of follicle to form into corpus luteum
Gonadotropin-releasing Hormone (GnRH)	Hypothalamus	Ovulation	Triggers anterior pituitary gland to release FSH and LH
Relaxin		Whole pregnancy until labour	Inhibits uterine contractions to prevent premature birth

Labour and Birth in Humans

- For baby to be born:
 - Muscles in the uterus must contract
 - Cervix must soften to allow passage of the baby
- Prostaglandin are secreted by the wall of the uterus & cause the tissue to be more sensitive to oxytocin → promotes the contractions of the smooth muscle in the cervix
- Oestrogen & Progesterone decrease during labour
- After birth – oxytocin continues to be secreted to allow the placenta to be expelled
- Prolactin is produced by the pituitary gland & stimulates milk production

Manipulation of Reproduction in Agriculture

- Mass artificial pollination in plants → remove pollen distribute to female flower to fertilise
 - Adv: desirable traits, Disadv: costly
- Cloning → copies of genetic code and implanting in parent
 - Disadv: ethical issues, costly, decreased biodiversity
- Vegetative propagation → pieces of plant can be grown into a new plant
 - Used in agriculture to increase production of crops when seeds are unavailable/difficult to germinate (eg. watermelons, mangoes)
 - Also used if plants have specific desirable traits that farmers want to perpetuate in future generations
 - Adv: cheap, simplistic, Disadv: decreased biodiversity
- Artificial insemination
- Selective breeding
- Genetic modification (AKA transgenic species)

Cell Reproduction

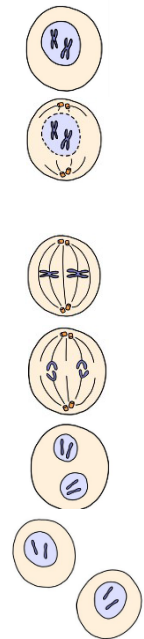
IQ 2: How important is it for genetic material to be replicated exactly?

- model the processes involved in cell replication, including but not limited to:
 - mitosis and meiosis
 - DNA replication using the Watson and Crick DNA model, including nucleotide composition, pairing and bonding
- assess the effect of the cell replication processes on the continuity of species

Mitosis

- Division of the nucleus as a single cell divides into two cells
- Purpose:
 - GROWTH – for multicellular organisms
 - REPAIR – damaged/worn out cells
 - GENETIC STABILITY – each cell has a full set of chromosomes
 - ASEXUAL REPRODUCTION – unicellular organism
- Stages of Cell Growth
 - G1 – gap/growth phase, prep for division
 - S – synthesis phase, DNA replicates
 - G2 – 2nd growth, checks DNA
 - M – mitosis/division

} interphase
- IPMAT
 - I (intro) – interphase
 - Growth stage of cell cycle
 - P (pair) – prophase
 - Chromosomes thicken, nuclear membrane breaks down, mitotic spindle begins to form, centrosomes move to opposite poles
 - M (middle) – metaphase
 - Chromosomes line up with centre of the cell, fibres attach to the centromere
 - A (apart) – anaphase
 - Fibres contract, pulling segments of the chromosomes in opposite ends of the cells
 - T (two) – telophase
 - Mitosis done, nuclear membranes form around chromosomes
- Mitosis ends with *cytokinesis*
 - As the nuclear membrane reforms in anaphase, the cytoplasm begins to be split by a cell membrane, separating the two newly formed daughter cells

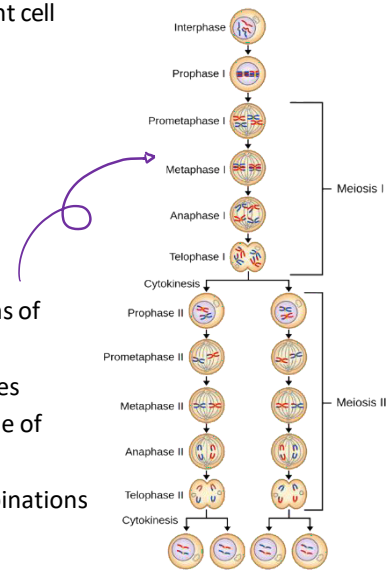


Limited Divisions – Cause of (Cell) Aging and Danger of Cancer Cells

- Aging → telomere shortening over time to the point where there isn't enough DNA and the cell dies
- Danger of cancer cells
 - Divide at faster rate than normal cells
 - Look different and may become less specialised
 - DNA mutates, making them resistant to earlier successful treatment
 - Can colonise new parts of the body and continue to grow unchecked
 - Can continue dividing endlessly (telomeres do not shorten)
 - Not affected by normal signals that control the cell cycle, eg. contact inhibition

Meiosis

- Haploid cells are created that are genetically different from the parent cell
- "Reduction division"
- Undergoes two divisions which creates 4 daughter cells
- Creates gametes (haploid) which, once they undergo fertilisation, combine to form a diploid
- Divisions:
 - Meiosis 1
 - Chromosomes replicate during this stage
 - Chromosomes may "cross over" exchanging lengths of matching DNA
 - Crossing over – homologous chromosomes exchange sections of DNA during prophase of meiosis
 - Creates an unlimited set of possible combinations which leads to genetic variation
 - Meiosis 2
 - No replication occurs here, before the second division, resulting in four daughter cells with only half genetic information

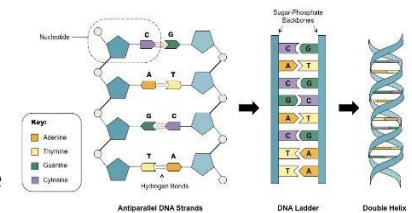


Comparison

Meiosis	Mitosis
Gametic cells	Somatic cells
Four daughter cells formed	Two daughter cells formed
Daughter cells have haploid cells (n)	Daughter cells have diploid cells (2n)
Necessary for sexual reproduction	Necessary for growth and repair
Crossing over takes place	No crossing over takes place

DNA Structure

- Located in nucleus
- Single strand tightly wound to form one chromosome
 - Instructions for our cells, so every cell needs a full set
- Model discovered by Watson & Crick through X-ray / chemical evidence
 - Identified shape as twisted ladder with pairs of bases
- Simply, consists of a repeating pattern which is a single unit (nucleotide) made up of three smaller components
 - Nitrogenous bases that join in pairs via hydrogen bonds
 - Adenine-Thymine
 - Guanine-Cytosine
 - Phosphate group
 - Sugar group (5-C pentagon) / pentose
- Each side runs in opposite directions



DNA Replication

- Occurs so an exact copy of the DNA molecule can be made, ready for cell division, without losing any of the genetic information
- DNA double helix unwinds
 - DNA unzips (by enzyme DNA helicase)
 - Nucleotides are added to each single strand by matching base pairs (by enzyme DNA polymerase)
 - Incorrect base pairing will result in a change in the DNA base sequence (mutation)

- Results in 2 molecules of DNA identical to each other and the original molecule from which they were formed
- Needed for heredity and gene expression (protein synthesis, which relies on DNA replication)

Chromosomes

- Compact coils of DNA, linear sequence of genes
- Each gene is made of a portion of DNA, with each gene located at a particular site (locus) on the chromosome
- Chromatid:
 - Single strand of DNA
 - During cell division, DNA is replicated where the resulting two identical copies of DNA are connected to each other at the centromere
 - Each copy is called a chromatid
 - A chromosome before cell division is formed by one chromatid, while it is formed of two chromatids after cell division

Other Definitions

- Genome – complete set of genetic information in an organism
- Phenotype – observed physical expression of an organism
- Genotype - Information contained in your DNA
- Mutations – change in sequence of bases (eg. deletions, inversions)
 - May result in new allele = new genetic code = new polypeptide = new phenotype = variation

DNA and Polypeptide Synthesis

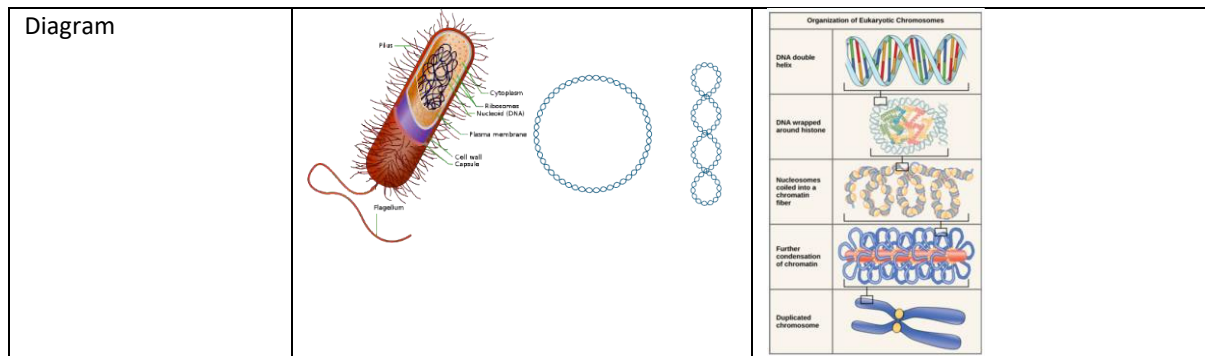
IQ 3: Why is polypeptide synthesis important?

- construct appropriate representations to model and compare the forms in which DNA exists in eukaryotes and prokaryotes
- model the process of polypeptide synthesis, including:
 - transcription and translation
 - assessing the importance of mRNA and tRNA in transcription and translation
 - analysing the function and importance of polypeptide synthesis
 - assessing how genes and environment affect phenotypic expression
- investigate the structure and function of proteins in living things

DNA in Eukaryotes and Prokaryotes

- Genetic code is universal – same nucleotide base pairing is used in all organisms
- There are differences at the level of gene expression – how instructions in DNA are converted to products like protein

	Prokaryotes	Eukaryotes
Location and Structure	1 chromosome – circular strand of DNA (called a <i>genophore</i>) No nucleus, floats in cytoplasm	DNA arranged into a number of linear, separated chromosomes Located in nucleus
Packaging	1300um in length so supercoiled to fit inside 3um organisms	Linear winds around proteins (histones) but not supercoiled
Other DNA	<i>Non-chromosomal DNA:</i> May have 1+ non-chromosomal DNA (plasmids) floating separately in the cytoplasm Provide selective advantage to the organism (can pass traits within generations) Replicate independently of the chromosome	<i>Non-nuclear DNA:</i> Mitochondria and chloroplasts also contain DNA Inherited

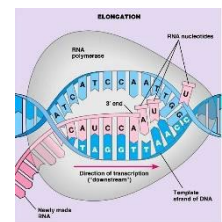


RNA

- Like DNA, RNA is a nucleic acid made up of a chain of nucleotides but differs in:
 - Mostly single stranded
 - Sugar in RNA is a ribose sugar (not deoxyribose sugar as in DNA)
 - Uracil (U) instead of thymine (T)
- 3 types:
 - Messenger RNA (mRNA): found both in nucleus and cytoplasm, functioning as an intermediate molecule, carrying information from DNA in the nucleus to the ribosomes in the cytoplasm
 - Transfer RNA (tRNA): cytoplasm, one end three unpaired bases (anticodon), the other end able to bind temporarily to an amino acid, each tRNA only attaches to one particular amino acid
 - Ribosomal RNA (rRNA): forms a structural part of ribosomes and made in nucleolus of cell

Protein Synthesis

- Proteins:
 - Large macromolecules made up of 1+ chains of polypeptide
 - Sequence of amino acids determines the polypeptide formed
 - Sequence of amino acids is determined by sequence of base pairs on the DNA molecule
- Steps (transcription → transfer → translation)
 - DNA unzips (RNA polymerase – enzyme which helps make mRNA)
 - Transcription occurs (DNA acts as a template for the formation of a complementary strand of messenger RNA (mRNA))
 - Here, Thymine is replaced by uracil
 - RNA is $\frac{1}{2}$ strand of DNA
 - Transfer: mRNA moves from nucleus into cytoplasm, and moves to ribosomes
 - mRNA rather than DNA as it is small enough to move through pores in nuclear membrane (DNA is complex and very valuable)
 - Translation
 - Ribosomes move along the mRNA molecule and attach tRNA molecules by pairing the bases of tRNA anticodons with complementary bases on mRNA
 - Anticodon - trinucleotide sequence located at one end of a transfer RNA (tRNA) molecule
 - Amino acids are sliced off their tRNA carriers and are linked to form a polypeptide chain
 - Each triplet of nucleotide bases codes for a different amino acid
 - tRNA moves out and into cytoplasm to pick up another amino acid
 - Chain of amino acids become a polypeptide chain
 - Chains of polypeptides are converted into proteins



MORE DETAILED

Transcription:

- Begins when RNA polymerase binds to a section of DNA and begins building a chain of RNA molecules to form a complementary strand of RNA
- 1. RNA polymerase binds to DNA and DNA 'unzips' [DNA un-spirals, hydrogen bonds between the two strands break, strands separate over a short length], only in the area where the gene is to be used
 - a. Only one strand of DNA contains the genetic information to make a protein (non-coding strand/sense strand)
- 2. Transcription controlled by RNA polymerase – the sense strand of the DNA acts as a template and RNA nucleotides are assembled, forming a complementary single-stranded mRNA molecule, also where T is replaced by U

Transfer:

1. mRNA, now small enough to move through pores of nucleus, exits nucleus into cytoplasm where it encounters some of the millions of ribosomes in the cell
 - a. Usually one mRNA molecule is read by a large number of ribosomes, so multiple chains of the same polypeptide product are produced from one mRNA template molecule

Translation:

1. Occurs when the ribosomes move along the mRNA molecule and attach tRNA molecules to mRNA by temporarily pairing the bases of the tRNA anticodons with their complementary triplets of bases (codons) on the mRNA
2. Amino acids from the tail end of each tRNA are linked to one another by an enzyme to form a polypeptide chain
3. Each amino acid is then spliced off its tRNA carrier
4. tRNAs move away from the mRNA, leaving the growing chain of amino acids and move back into cytoplasm where they can pick up another amino acid and be reused

Function and Importance of Polypeptide Synthesis

- Vital for producing proteins
- Proteins are necessary for controlling and coordinating structure and function within cells

Structure and Functions of Proteins in Living Things

- Chemical
 - CHON combine to form amino acids, held together by peptide bonds
- Physical
 - Primary:
 - Polymer of amino acid arranged in linear chains/polypeptides
 - Secondary:
 - Local folding of polypeptide chain, due to hydrogen bonds, twisting into alpha helix or folding into pleated sheets
 - Tertiary:
 - Further folding of secondary proteins
 - Certain forces of attraction due to side chain interactions between alpha helices of pleated sheets cause polypeptide to fold into more complex 3D shapes
 - Quaternary:
 - 2+ polypeptide chains that link to create more complex 3D structure
- Types of Proteins
 - Fibrous → structural components of cells and tissues
 - When combined with water, form basic structure of protoplasm eg. collagen
 - Long, insoluble
 - Globular proteins → spherical shape that transport protein
 - Soluble in water, compact

- Functions
 - Form structural and functional part of cell membrane
 - Communication and regulate movement of substances across cell membrane
 - Motility (eg. myosin and actin) -> join to form filaments and that allow movement

Protein Example – Haemoglobin

- Structure: comprises of 4 subunits, each having 1 polypeptide chain, 2 alpha chains and 2 beta chains, each with a ring-like heme group containing iron atom
- Function: two-way respiratory carrier, carrying O₂ from lungs to the tissues and facilitating return of CO₂
- Structure enables function: each iron in each heme group can bind to one oxygen, therefore each haemoglobin molecule can bind to 4 oxygen

Protein Example – Amylase

- Structure: Tertiary protein with complex folds
- Function: Digestive enzyme that catalyses the breakdown of starch into maltose
- Structure enables function: 3D structure and highly folded nature increases surface area to increase rate of catalysation

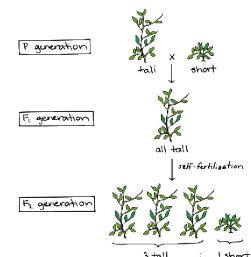
Genetic Variation

IQ 4: How can the genetic similarities and differences within and between species be compared?

- conduct practical investigations to predict variations in the genotype of offspring by modelling meiosis, including the crossing over of homologous chromosomes, fertilisation and mutations
- model the formation of new combinations of genotypes produced during meiosis, including but not limited to:
 - interpreting examples of autosomal, sex-linkage, co-dominance, incomplete dominance and multiple alleles
 - constructing and interpreting information and data from pedigrees and Punnett squares
- collect, record and present data to represent frequencies of characteristics in a population, in order to identify trends, patterns, relationships and limitations in data, for example:
 - examining frequency data
 - analysing single nucleotide polymorphism (SNP)

Alleles and Autosomal Inheritance

- Allele – alt form of a gene (eg. eye colour)
 - Occur in pairs (one from each parent)
 - Chromosomes come in pairs, each contain an allele for every gene
- Autosomal Inheritance – refers to genes that code for characteristics in body cells (ie. not sex cells)
- Gregor Mendel
 - Using pea plants and artificial pollination techniques, he completed detailed experiments regarding somatic inheritance
 - Each feature is controlled by 2 factors (inherited from one's parent)
 - Some traits can be hidden during a generation (recessive)
 - He took a tall plant and bred it with a short plant where the offspring were all tall
 - When he crossed plants from this new generation, he produced 3 tall offspring and 1 short



- Heterozygous condition: chromosomes have different alleles of the same gene (eg. T and t)
 - Recessive gene is turned off, and dominant gene shows in phenotype
- Homozygous dominant condition: both chromosomes have the dominant allele (eg. T)
- Homozygous recessive condition: both chromosomes have the recessive allele (eg. t)

Punnett Squares

- Simple tool to estimate possible outcomes with a mating, showing different combinations of gametes being produced. (eg. pea plants above can be demonstrated using punnett squares)

Sex Linkage

- Discovered by Thomas Morgan using fruit flies
 - Noted different coloured eyes (white) being more common in male flies
 - Gene for this trait is on the sex chromosome
- Sex chromosome = 23rd pair of human chromosomes (XX = F, XY = M)
- Fathers pass sex-linked alleles to all daughters, not sons
- Mothers can pass sex-linked alleles to daughters and sons
- Females: sex-linked recessive traits will be expressed only in the homozygous condition
- Males: express any recessive allele from their mother
- Eg colour blindness
 - Connected to the X chromosome

	Female	Male
Normal Vision	$X^B X^B$	$X^B Y$
Colour blind	$X^b X^b$	$X^b Y$
Carrier	$X^B X^b$	N/A as the father can only pass the X chromosome to females (the Y chromosome is too small to have any allele attached to it)

Co-Dominance

- No clear dominant feature, ie. both features are 'dominant' traits
- Both traits are visible in a heterozygous individual, in a third phenotype
- Eg. cattle coat colour controlled red and white gene (red coat = R, white coat = W)
 - Red coat genotype RR
 - White coat genotype WW
 - Heterozygous individual is described as roan

Incomplete Dominance

- Neither trait is visible in a heterozygous individual, a third trait is produced
- Eg. crossing a red flower with a white plant would give 100% offspring with pink flowers

Multiple Alleles

- More than two simple alleles, each with different levels of dominance and recessive
- Eg. Human blood has three alleles (A, B, O)
 - A and B dominant over O
 - A and B co-dominant
 - Form 6 genotypes and 3 phenotypes

Pedigree Trees

- Visual representations of how phenotypes are passed along generations within a family
 - Can then be used to figure out genotypes
- Trace inheritance patterns of disorders or breeding
- Used to determine if a trait is dominant, recessive, sex-linked or co-dominant
- M = squares, F=circles

Population Genetics

- Study of how a gene pool (sum of total genes) of a population changes over time, leading to species evolving
 - Species with a greater degree of genetic diversity have a greater potential to adapt and survive
- Frequency studies are conducted to predict resilience, likelihood of survival and ability to adapt

Qualitative Analysis

- Trend: A tendency of a set of data to follow a certain direction
- Pattern: when data is repeated, it is following a particular way/trend
- Relationship: exists between two or more variables where one is changed, it influences the other

Quantitative Analysis – Allele Frequency

- Useful: Apply mathematical models to predict how external factors will influence these frequencies, genetic variability in a population can be determined by analysing the relative proportion of a given phenotype, genotype or allele
- Allele frequency – measurement of how common a specific allele is, impacted by surrounding environment
- Allele frequency studies – calculate the percentages of certain alleles/genotypes/phenotypes to see how they change across generations, percentages need to add to 1
 - Freq of allele 'G' = no. allele G's in pop./Total no. of alleles

Single Nucleotide Polymorphisms (SNP)

- Mutation, where there is an error with a single nucleotide which must occur in at least 1% of the population
- Haplotype – group of SNP markers that are associated with a trait
 - Help to look at relatedness (evolutionary studies), family lineage and indicators of disease
- Types of SNPs
 - Linking SNPs – do not reside within genes and do not affect protein function, respond to a particular drug response or to the risk of getting a certain disease
 - Causative SNPs – affect the way proteins function, correlating with a disease or influencing a person's response to medication, come in two forms:
 - Coding: located within coding region of gene, change the aa sequence of the gene's protein product
 - Non-coding: located within the gene's regulatory sequences, change the time, location or level of gene expression
- Uses in genetic studies:
 - Important genetic marker (can tell individuals apart) which can be associated with specific traits or disorders, but do not necessarily cause them
 - Potential association with diseases and disease/traits that indicated
 - Genome mapping/recognising haplotypes
 - Haplotype – a group of SNP markers that are associated with a trait, which help look at relatedness (evolutionary studies), family lineage and indicators of disease
 - Limitations:
Haplotypes are more accurate when they are closer together; the further they are apart the more chance that crossing over could have occurred which means linking it to one parent is limited

Short Tandem Repeat (STR)

- Single mutation or locus (physical part of chromosome) repeats
- Unique to each individual

Patterns in a Population

IQ 5: Can population genetic patterns be predicted with any accuracy?

- investigate the use of technologies to determine inheritance patterns in a population using, for example:
 - DNA sequencing and profiling
- investigate the use of data analysis from a large-scale collaborative project to identify trends, patterns and relationships, for example:
 - the use of population genetics data in conservation management
 - population genetics studies used to determine the inheritance of a disease or disorder
 - population genetics relating to human evolution

Population Genetics

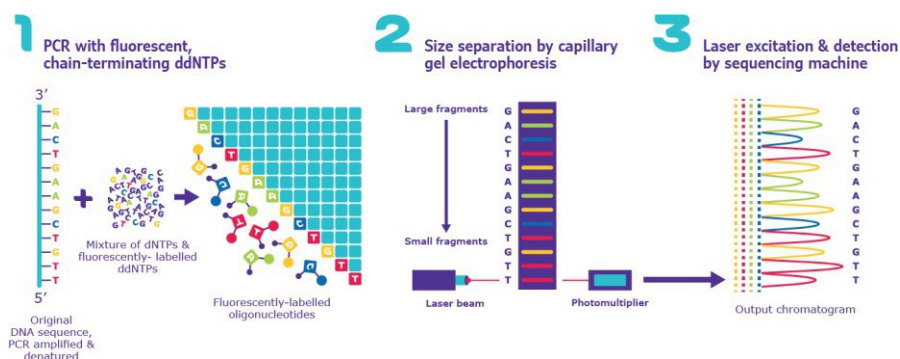
- Study of genetic variation within a population and the changes in the freq. of genes/alleles within a population and among populations over time

Human Genome

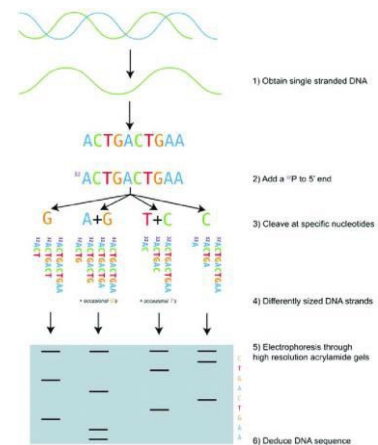
- Genome – ‘map’ of an organisms DNA
- 2003 & 2008 – human genome was mapped: Human Genome Project
 - Took 1000’s of people DNA from 20 universities and private companies across 13 years (1990-2003)
 - Mapped 3.2 billion bases in our haploid cells
- Genetic testing – simple medical test to detect alleles, mutations or genotypes associated with heritable traits, diseases or being predisposed to certain diseases

DNA Sequencing

- Studying of bases with a specific gene to learn the sequence of bases for one gene
- Specifically maps nucleotides of a DNA fragment, usually a gene
- General procedure involves amplifying a gene using a PCR from which they cut it and insert a fluorescent dye for a specific nitrogen base to tag that base
 - Then runs through a gel to learn where specific bases are
- Two main, explicit methods:
 - Sanger Method:
 - DNA is isolated and sorted into an order of length
 - DNA is divided in single strands with an added primer strand
 - End base is tagged with dye
 - Using an electric current, the DNA strand passes through gel and then through a laser detector
 - With the diff. lengths of DNA and differently dyed base tip, the unique properties of a species’ DNA can be visualised – useful for researching similarities and differences between diff. species



- Maxam-Gilbert Method:
 - Similar to Sanger but uses a radioactive phosphorus atom to tag/label the end of the DNA
 - Chemicals cleave/cut the DNA at certain bases with the diff. lengths passing through the electrophoresis gel



- Used to help us recognise STRs, where SNPs are located and to map a person's genes
- Difference between genotyping and sequencing:
 - Genotyping – identifying genetic variations in individuals
 - Sequencing – identifying the exact nucleotide sequence of a certain DNA length, more specific

DNA Profiling

- Identifies unique genetic markers, like STRs, unique to individuals
- Unique DNA sequence determined by the presence and location of STRs and SNPs
- Used in paternity and criminal cases, as a method of identification
- Identifies an allele rather than a nucleotide (easier than sequencing as greater scale)

Gel Electrophoresis

- Electric current passed through gel that is injected with dyed amplified DNA
- DNA separated by size/molecular weight
- Smaller DNA fragments travel further through the gel
- Thick band = homozygous (as same size)
- Thin, separated bands = heterozygous

Bioinformatics

- Study of genetic data analysis on a large scale

Population Studies and Conservation

- Conservation genetics = how genes are inherited in a population, in showing that some alleles are selected as they confer greater survival advantage than others
- Looking through DNA analysis aids our understanding of microevolution by studying mutation rates and how introducing certain individuals may help a population recover by increased genetic variation
- Alleles with negative impacts may be removed from the population
- Woolly Mammoth:
 - Study of these mammals' demise has allowed scientists to observe how any reduction in genetic diversity may have accelerated (or slowed) the extinction of these creatures
 - Comparing Wrangel Mammoths (an isolated species on Wrangel island) and mainland:
 - Genetic sequencing has told scientists various genetic errors in island's population including poor olfactory senses, urine problems (causing mating issues), light shiny coats (causing issues with ice age)
 - Population plummeted from 10,000s to under 1,000
 - Isolation=Lack genetic diversity=extinction
- Koalas:
 - Control region of mitochondrial DNA in koalas was analysed to determine the diff. lineages and genetic diversity in koalas in eastern Aus
 - Recent genetic data have also provided statistics that cover a large area and a large timescale
 - DNA samples were also collected from 650+ koalas, looking for haplotypes (matching SNPs) which reflect relationships and the effect of mutations on a population

Importance of Genetic Variation

1. Increase gene pool = increased allele freq.=decreased susceptibility to certain diseases and environmental changes
2. Decrease chance of inbreeding and purebred traits
3. Increased hybrid vigour

Population Studies and Inheritance of Diseases

- Alkaptonuria (black urine disease):
 - Body can't process certain amino acids
 - With DNA sequencing, science has determined that this disease is due to a mutation in the HDG gene (monogenic disease)
 - These types of SNPs can be genetically tested for – this specific one, at birth through a heel prick
- In NSW, newborns are genetically tested for a no. of congenital diseases (existed at birth) with the hope of creating a database for gene-specific disease mutations (longitudinal study)

Population Studies and Human Evolution

- Anthropology (study of human evolution) generally based on fossil record and physical comparisons, but with rapid advances in DNA technology, we can bring in a greater diversity of data including SNP's, allele freq. and genetic sequencing
- Human Migration – using genetic markers we can trace how/where humans have evolved:
 - Multiregional Hypothesis: relies mostly on fossil record and suggests that all human populations can be traced to when Homo erectus first left Africa, suggested gene flow between neighbouring populations and that once that dispersed into other portions of the old world, they slowly evolved into modern humans
 - Replacement Hypothesis: second migration out of Africa and that modern humans of African origin conquered archaic groups and replaced them by interbreeding with and out-competing them
- Can trace two pieces of evidence:
 - Mitochondrial DNA (maternal trace – mitochondria sits in egg)
 - Y-chromosome trace (male)
- Indigenous Australians
 - Due to isolation, genetically we see some unique DNA markers from the Denisovan subspecies

Module 6: Genetic Change

Mutation

IQ 1: How does mutation introduce new alleles into a population?

- explain how a range of mutagens operate, including but not limited to:
 - electromagnetic radiation sources
 - chemicals
 - naturally occurring mutagens
- compare the causes, processes and effects of different types of mutation, including but not limited to:
 - point mutation
 - chromosomal mutation
- distinguish between somatic mutations and germ-line mutations and their effect on an organism
- assess the significance of 'coding' and 'non-coding' DNA segments in the process of mutation
- investigate the causes of genetic variation relating to the processes of fertilisation, meiosis and mutation
- evaluate the effect of mutation, gene flow and genetic drift on the gene pool of populations

Mutation

- Any change in a genetic sequence involving 1+ nucleotides
- May result in a beneficial or harmful phenotype or no change at all depending upon the location along the DNA

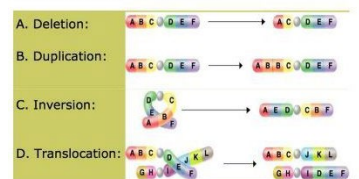
Point Mutations

- Occur along the DNA and only impact a small number of nucleotides
- Substitution – one base is replaced by the wrong match:
 - Silent: despite a change in one nitrogen base, the amino acid is not changed, therefore phenotype is not affected
 - Missense: a change in nitrogen base results in the change of one amino acid in the polypeptide
 - Nonsense: premature stop codon which shortens the polypeptide and so usually results in a non-functional protein
- Frameshift mutation – nucleotide is added or deleted which in turn impacts all the following codons, and thus amino acids being coded for

Chromosomal Mutations

- Occurs during meiosis which impact large sections of DNA/entire chromosome
- Duplication:
 - Segment (few nitrogen bases to whole chromosome) is produced
- Deletion:
 - Loss of up to one whole chromosome
- Inversion:
 - Segment of DNA breaks away and reattaches in reverse order
- Translocation:
 - Piece of chromosome breaks off and attaches to another
- Aneuploidy: 1+ chromosomes missing or in surplus
 - Eg. Down Syndrome, Trisomy (3 chromosomes instead of 2 on chromosome)
- Polyploidy: An extra complete set of chromosome
 - Common in plants, salamanders, frogs, leeches

Chromosomal Mutations



Mutagens

- An agent that causes a genetic mutation
- Chemical:
 - Chemicals that cause mutations if exposed to for long periods of time

- Often similar to structure of DNA, they can be mistakenly incorporated into the DNA and cause issues with protein synthesis
- Ingested: alcohol, cigarette smoke, medications, fatty foods, additives, preservatives
- Environmental: pesticides, lead, benzenes
- Naturally Occurring:
 - Occur in nature, at levels in the environment that would cause mutations. If levels increase, likelihood of mutations occurring would also increase
 - Biological: viruses, bacteria, fungi
 - Non-biological: heavy metals (ie. lead, nickel, arsenic)
- Biological:
 - Microbes (eg. HPV) that cause mutations in human genes
 - Retroviruses insert their DNA in the host's chromosomes causing some types of leukaemia
 - Some mutagens formed when metabolising certain foods -> cooking/frying certain combinations can create carcinogens
 - Transposons (transposable elements TE) are short segments of DNA that move about the human genome which can disrupt protein synthesis if they enter a gene segment (eg. haemophilia)
- Physical (non-biological)
 - Eg. ionising radiation where radiation possesses enough energy to push an electron out of its atomic orbit, breaking bonds and damaging the DNA molecule
 - Eg. UV rays in range of EM which changes between ionising and non-ionising
 - UVA has longest wavelengths (315-400nm) and relates more with premature ageing than ionising affect
 - UVB (280-315nm) & UVC (180-280nm) have shorter wavelengths and carry more energy, meaning they can cause damage to the cells and DNA they contain; causes T bases to join together

Somatic VS Germ-line Mutations

- Somatic:
 - Occur in body cells, most often to replication errors prior to mitosis.
 - Spontaneous mutations may occur in the S phase of the cell cycle and, if not repaired during 'proofreading' in the G2 phase, will be passed on to daughter cells
 - When a mutated cell continues to divide by mitosis, the error is replicated each time and passed on to cells in successive divisions, amplifying the error within that tissue
 - Do not always result in visible phenotypic changes in cells
 - Earlier the mutation occurs, the greater its effects will be as the mutation will be replicated in a greater number of cells as the organism grows
- Germline:
 - Occur in the sexual reproductive cells that give rise to gametes
 - Passed to offsprings
 - When a gamete carrying the mutation fuses with another gamete and an embryo forms, the mutation is replicated in every cell of the embryo as it divides and grows, affecting all cells in the resulting child

Coding Regions and Non-coding Regions

- Most of our DNA is not used for protein synthesis ('junk DNA')
- Exons – lengths of DNA that carry a gene, which codes for polypeptide synthesis (coding segment)
 - Mutations occurring within an exon:
 - Change/alt. DNA (aka potentially lead to a change in a.a -> leads to a change in polypeptide -> change in phenotype/genotype)
 - Could be a lack in functioning within the body
- Introns – sections of DNA do not code for a.a and polypeptides (non-coding segment)

- Mutations occurring within an introns:
 - While do not directly code for polypeptides, they often can contain code for promotion and termination patterns
 - They contain a base sequence to regulate translation or transcription
 - If an error occurs in these areas of DNA, then the result can lead to a serious genetic disorder, eg. CF

Causes of Genetic Variation

- Fertilisation and GV
 - Increase in GV as there is $\frac{1}{2}$ set of chromosomes (haploid) coming from each parent
 - Mix of paternal and maternal alleles
- Meiosis and GV
 - Crossing over further increases GV as alleles are swapped between homologous chromosomes
 - Creates recombination of genotypes
 - Independent assortment (how they line up in metaphase) -> increase in GV as bivalent pairs line (independent of the other chromosome)
 - Random segregation -> splitting haploid daughter cells in final stage
- Mutation and GV
 - Any mutation has potential to create new alleles -> increase in alleles for a particular allele = increase GV of a gene pool

Gene Pool, Gene Flow and Genetic Drift

- Gene pool – total amount of genetic variation within a population of a species. Measured by allele and genotype frequency
 - Can change by: mutation, gene flow, genetic drift, natural selections
 - Mutations: increase gene pool if they are beneficial for the individual, lethal mutations will make no long-term change to a gene pool (won't survive = won't pass on traits)
- Genetic Drift:
 - Random change in allele frequency by chance (unlike natural selection which is not based on chance but on favourability of alleles)
 - "Founder effect" – small group separated from larger groups eg. island = decrease variation due to lower number of alleles
 - "Bottleneck effect" – when a disaster eg. bushfire reduces population size -> reduces allele variation -> more susceptible to further reduction of alleles as less hybrid traits
 - Eg. 1890s, elephants seals were hunted down to the last 20 individuals. Now at 30,000 individuals, still very little GV (last 20 by chance)
- Genetic Flow:
 - Passing on genetic material from one population to another
 - Selects traits
 - Can either increase or decrease GV
 - Eg. migration of animals into a population = individual new alleles = increased gene pool

IQ 2/3 blend together:

Biotechnology

IQ 2: How do genetic techniques affect Earth's biodiversity?

- investigate the uses and applications of biotechnology (past, present and future), including:
 - analysing the social implications and ethical uses of biotechnology, including plant and animal examples
 - researching future directions of the use of biotechnology
 - evaluating the potential benefits for society of research using genetic technologies
- evaluating the changes to the Earth's biodiversity due to genetic techniques

Definition [examples in IQ 3]

- Process of utilizing biological agents/systems to develop/create different products
- Around for 10,000s of years – basically whenever humans have purposefully manipulated biology for their gain
- Branches: agriculture, medicine and health, industrial processes involving microorganisms, food and nutrition, environmental biotechnology, bioinformatics, ethics and philosophy, bioterrorism

Biotechnology Past, Present and Future – Uses and Applications

- Past:
 - 9000 years ago in Mesoamerica: selectively breeding (application) teosinte plants that had larger kernels (use), leading to development of 1st corn species
- Present:
 - IVF (in vitro fertility): process by which an egg is combined with sperm in vitro (application); involves monitoring and stimulating a woman's ovulatory process, removing an ovum from the ovaries and letting a man's sperm fertilise them in a culture medium in a laboratory – used to select for healthy, advantageous traits or to eliminate unhealthy traits or allows infertile individuals to procreate (uses)
- Future:
 - CRISPR (Look at IQ3), nanotechnology eg. quantum dots for medicine, therapeutic cloning (Look at IQ 3)

Social Implications and Ethical Uses (broadly)

- Social – affect the way people live/work in their environment
- Ethical – concerned with what is right/wrong

Ethics/Social Implications	
<u>Positive</u>	<u>Negative</u>
Faster growth rate	Harm to Environment
Can create extended life	Bioterrorism
Specific, beneficial traits can be developed	Affordability
New products can be created	Harm to animal/human subjects in clinical trials
Greater yields can be produced	Laboratory/production safety

Ethical Questions

- Benefit/Non-harm: whose best interests is the action (individuals, group, society, future generations), reliability, do advantages outweigh the risks
- Individual rights and autonomy: person properly informed of advantages/risks, does the subject have freedom of choice, are rights of others protected/breached
- Privacy/societal perception: will the person's genetic information remain private, how might the release of this information affect the way in which society/another individual perceives the person
- Equity/justice: equal access for all members of society, can discrimination be avoided, fair sharing of resources that can be justified

Changes to Biodiversity and Genetic Technologies

- Biodiversity = 'variation' within life
 - Genetic diversity: variety of genes/alleles within a species
 - Species diversity: variety of species in a particular habitat
 - Ecosystem diversity: variety of ecosystems within an area
- Genetic Technologies and impact on biodiversity:

- Can introduce a gene into an organism to create a phenotypical change, eg. Genetically modified salmon aka Aquadvantage salmon:
 - Insertion of a growth gene from chinook salmon and eels, meaning that they can grow 3x as large, have more flesh for eating and reach this size much quicker
 - This has great economic benefit
 - However, it poses a threat to natural wildtype population – if the Aquadvantage salmon escaped into the wild, it could alter the food chain, form competition within existing salmon and through breeding, alter the gene pool
 - Scientists are ‘shocking’ fertilised GM eggs so the adults are infertile and therefore can not pass on transgenic trait
- Concern of GM species is that they could ‘contaminate’ the wild population
- Conservation genetics mean endangered animals are less likely to become extinct – DNA sequencing and profiling can detect low genetic diversity which allows for scientists to intervene to increase geneflow or use specific breeding program to increase the genetic diversity within a population (therefore increasing biodiversity)
- Humans can now alter evolutionary pathways, combining separate species
 - Humans will breed and reproduce only individuals with features preferable to our needs and not evolutionary needs of the organism, leading to a decrease in biodiversity
- Introducing new genomes will increase biodiversity in the short term

Genetic Technologies

IQ 3: Does artificial manipulation of DNA have the potential to change populations forever?

- investigate the uses and advantages of current genetic technologies that induce genetic change
- compare the processes and outcomes of reproductive technologies, including but not limited to:
 - artificial insemination
 - artificial pollination
- investigate and assess the effectiveness of cloning, including but not limited to:
 - whole organism cloning
 - gene cloning
- describe techniques and applications used in recombinant DNA technology, for example:
 - the development of transgenic organisms in agricultural and medical applications
- evaluate the benefits of using genetic technologies in agricultural, medical and industrial applications
- evaluate the effect on biodiversity of using biotechnology in agriculture

interpret a range of secondary sources to assess the influence of social, economic and cultural contexts on a range of biotechnologies

Artificial Pollination VS Artificial Insemination

- Artificial Pollination: mechanical technique completed by hand, commonly with a brush or a cotton swab, used to pollinate plants to achieve desirable traits; involves distributing pollen from a desirable male anther of a plant to the female stigma of another plant
- Artificial Insemination: deliberate insertion of male sperm directly into the cervix or uterus of a female of the same species, close to ovulation, to achieve pregnancy. It is used both in agriculture for desirable stock and in humans.

Similarities	Differ in Outcomes:
Require human intervention	Artificial pollination usually conducted in order to produce a high yield plant with desirable traits
Unnatural processes	However, artificial insemination is completed to create offspring with more specialised traits by isolating parent gametes with desired traits
Conducted in controlled environments	

Involve processes of selectively choosing desirable males gametes to insert in female, to produce selectively bred individuals	
Male gametes can be cryopreserved for future use	

Effects of IVF

- Genetic Diversity: possibly reduced due to multiple viable embryos supplied by a small selection of parents with favoured features
- Genes for infertility: infertility would naturally disappear from a population however, with IVF, humans can now breed infertility into the population
- Sperm Banks: may alter gene frequency with people selecting donors with features perceived as favourable

Effect of Biotechnology on Biodiversity

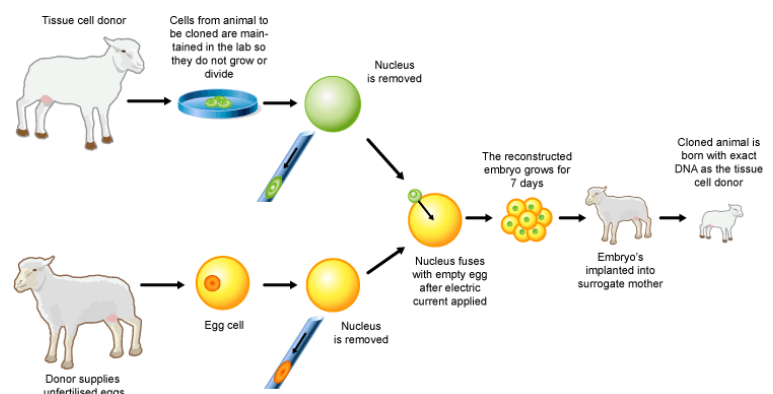
- Wildlife Conservation: Focus is to save/rescue endangered/extinct species -> we want to increase GV in order to help endangered species survive -> by increasing GV, hybrid vigour is increased therefore increased chance of species surviving
- Agriculture: Focus is for crops/cattle to have specific, favourable traits for high yield (=high profit) and good health, so GV will reduce to make desired feature(s) more common

Cloning

- Creating a genetically identical copy of a gene, cell or whole organism

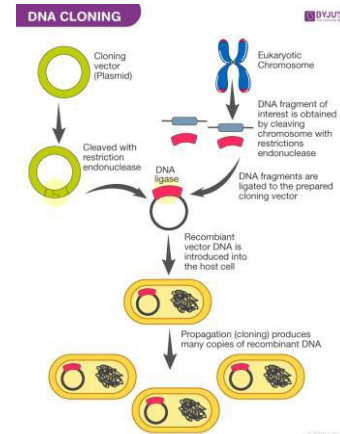
Whole Organism Cloning

- Form of reproductive cloning where a whole new, genetically identical individual is produced via SCNT (somatic cell nuclear transfer)
- Process (SCNT):
 1. Isolate donor cells – somatic cell from individual with desired genetics and germline cell (egg)
 2. Remove and discard nucleus from the egg. Remove nucleus from somatic cell.
 3. Transfer (via needle) the somatic cell nucleus into the enucleated egg cell
 4. Stimulate cell division
 5. Implant the embryo into surrogate mother
 6. Deliver baby (baby is genetically identical to somatic cell donor)
- Can clone via twinning (embryo splitting)
 - Simpler process where a female individual is treated with hormones to cause superovulation and produce many eggs -> ovum are fertilised, flushed out of mother and allowed to divide (via mitosis) -> divided cells are split into smaller embryo which are grown in the lab before being transferred into a surrogate mother
 - Each cell will create individuals genetically identical to mother
- Inefficient, eg. it took 276 attempts to clone Dolly the Sheep and has less than 1% success rate. It doesn't really help society on an everyday (medical) basis though the potential for whole organism cloning would be beneficial for species conservation (endangered and extinct species)



Gene Cloning (Recombinant DNA)

- Specific copies of a DNA segment (gene) are cloned
- 1. Restriction digestion (cutting) of a vector (any particle used as a vehicle to artificially carry a foreign nucleic sequence – usually DNA – into another cell, where it can be replicated and/or expressed eg. bacteria plasmid)
- 2. Restriction digestion (cutting) of a gene of interest
- 3. Ligation of gene into plasmid (insert gene of interest into plasmid) with DNA ligase (an enzyme) which joins the fragments with matching ends together to form a single, unbroken molecule of DNA -> this ligation creates a recombinant plasmid
- 4. Transformation of the recombinant plasmid into competent bacterial strain
- 5. Growing the bacteria in bulk amount to amplify gene of interest (giving it its optimum environment)
- Highly efficient (high yield for relatively low cost/resources) and produces consistent results with great yields
- Highly beneficial to amplify desire genes, eg. Humulin – insulin taken from bacteria/pigs to produce insulin for diabetes patients



Therapeutic Cloning (outside syllabus – good future example for biotechnology)

- Cloning designed as therapy for a disease whereby the nucleus of a somatic cell is inserted into a fertilised enucleated egg -> enucleated egg begins to divide repeatedly to form a blastocyst from which stem cells can be extracted to use to grow cells that are a perfect genetic match for the patient -> can then be transplanted into the patient to treat a disease

Recombinant DNA Technology

- Technology used for producing artificial DNA through the combination of different genetic materials from different sources
- Involves using enzymes and technologies to manipulate and isolate DNA segments of interest
- See Gene Cloning (IQ 3) for in depth
- 1. Isolation of Genetic Material -> desired DNA is isolated in its pure form
- 2. Cutting the gene at recognition sites -> restriction enzymes (isolated from bacterial cells) determine location at which desired gene is inserted into the vector genome
- 3. Amplifying gene copies through PCR -> single copy of DNA amplified into thousands of copies once the proper gene of interest has been cut using restriction enzymes
- 4. Insertion of recombinant DNA into host (transformation) -> once recombinant DNA is inserted into host cell, it gets multiplied and expressed in the form of the manufactured protein

Advantages	Disadvantages
Increased resistance of crop plants to harsher environmental conditions/pests	Ethics: Should mankind have and use the means to modify their own nucleic acids for enhancing themselves?
Greater innovation in the field of medicine to produce new medicinal tools/drugs to treat diseases	Equal access?
Genetic engineering of recombinant DNA to code useful proteins	Will underprivileged people become further disadvantaged without access to biotechnology products
Development and insertion of DNA therapeutics into damaged chromosomes to treat hereditary diseases	

Transgenic Species

- An organism whose genome has been altered by the introduction of 1+ foreign DNA sequences from another species
- This occurs when a gene from one organism is inserted into the genome of another individual of another species
- Eg. Aquadvantage salmon, anti-freeze strawberries, golden rice, Bt cotton (gene from bacteria to make cotton insecticidal) , Enviropig (altered so urine contains up to 65% less phosphorus (better for the environment) by adding phytase gene from E.coli bacteria causing the pig to produce its own phytase in its saliva and through digestive tract to break down phosphorus)

Ethics

Pros/Advantages	Cons/Disadvantages
Utilitarian (better for wider society than potential harm of one individual)	Animal rights -> violates autonomy of an individual animal
Increases agriculture efficiency (output>input) -> subsequently better for the environment	Unexpected side effects
Disease Resistance	"Playing God" -> religious consequences
Gene pharming for desirable traits to cure diseases, drug discovery and tracking	Unequal access

CRISPR-Cas 9 – Future

- Enables scientists to edit parts of the genome by removing, adding or altering sections of the DNA sequence
 - Involves cutting out a target DNA sequence via a system made up of two key parts: a CRISPR-associated nuclease, which binds and cuts DNA, and a guide RNA sequence, which directs the nuclease to its target
1. Gene being modified must be decided
 2. Appropriate endonuclease protein must be selected
 3. gRNA to target the gene of interest must be decided
 4. gRNA Expression Vector assembled, plasmid assembled
 5. New cells engineered

Advantages	Disadvantages
Higher gene editing efficiency	Potential for off-target genome editing effects
Lower off-target effect	May induce site-specific DNA mutations
No DNA integration	Expensive
Environmentally-friendly	Ethical concerns, eg. use on human life
Key technology in fighting disease (eg. malaria - reduces spread by gene editing malaria or making coronavirus tests)	

Benefits of Genetic Technologies in:

- Agriculture:
 - Can create crops better suited to unfavourable environments (ie. abiotic factors such as high salinity or drought)
 - Plants can be made to be resistant to pests (eg. Bt cotton)
 - Increase in productivity

- Producing GM plants with enhanced nutritional value (eg. Golden Rice with Vitamin A) or increased lipid content for increased oil production
- Medicine:
 - Cloning cells to help treat diseases such as Parkinson's and Alzheimer's
 - Processes like cloning, artificial insemination or IVF to either overcome infertility or to selectively pass on favourable traits
 - Artificial production of pharmaceutical products eg. Humulin – human insulin created by amplified bacteria (recombinant DNA)
 - Xenotransplantation – transplanting organs from one species to another with GM to ensure organs are not rejected (eg. pig hearts for heart disease)
- Industry:
 - GM foods used in food industry
 - Synthetic enzymes for textile productions and additives for detergents
 - CSIRO is developing GM potatoes to create a form of starch in paper and textiles
 - GM bacteria to help in heavy metal absorptions from contaminated sites

Module 7: Infectious Diseases

Causes of Infectious Disease

IQ 1: How are diseases transmitted?

- describe a variety of infectious diseases caused by pathogens, including microorganisms, macroorganisms and non-cellular pathogens, and collect primary and secondary-sourced data and information relating to disease transmission, including:
 - classifying different pathogens that cause disease in plants and animals
 - investigating the transmission of a disease during an epidemic
 - design and conduct a practical investigation relating to the microbial testing of water or food samples
 - investigate modes of transmission of infectious diseases, including direct contact, indirect contact and vector transmission
- investigate the work of Robert Koch and Louis Pasteur, to explain the causes and transmission of infectious diseases, including:
 - Koch's postulates
 - Pasteur's experiments on microbial contamination
- assess the causes and effects of diseases on agricultural production, including but not limited to:
 - plant diseases
 - animal diseases
- compare the adaptations of different pathogens that facilitate their entry into and transmission between hosts

Definitions

- **Disease** – a disordered or incorrectly functioning organ, part, structure or system of the body resulting from the effect of genetic or developmental errors, infection, poisons, nutritional deficiency, imbalanced toxicity or unfavourable environment - WHO
- **Health** – State of complete physical, mental, and social wellbeing, not merely the absence of disease or infirmity – WHO
- **Infectious disease** – a disease caused by a pathogen that can spread from person to person
- **Pathogen** – An agent, organism or microbe capable of causing a disease
- **Pathogenicity** – ability of a pathogen to cause a disease
- **Micro** – not visible to naked eye **Macro** – able to be seen with the naked eye
- **Epidemic** – Widespread occurrence of an infectious disease within a community at a particular time
- **Pandemic** – Widespread occurrence of an infectious disease over a whole country/large region (ie. the world) at a particular time
- **Zoonotic** – pathogens that can be transmitted between species
- **Transmission** – idea of a pathogen being spread/shared between different hosts
- **Virulent** – how easily spread a pathogen is

Types of Pathogens

Name	Cellular?	Description	Structure	Example
Prion	No	Abnormal protein that may cause degenerative diseases of the nervous system Smallest of all pathogens When in contact with 'normal' protein, they will change the structure into a new-prion protein	Contain no genetic material Protein-like structure	Creutzfeldt-Jakob disease (CJD) Mad Cow
Virus	No	Not able to reproduce, so require a host cell in order to create more viruses; - Virus injects its DNA/RNA in the host cell which then build new viruses	Single strand of DNA or RNA that is wrapped in a capsid protein coat	Influenza AIDS (HIV) Measles Covid-19

		- When host cell is 'full', it will erupt, releasing the virus and allowing it to infect new cells Retroviruses = contain RNA Bacteriophages = infect/invade bacterial cells		
Bacteria	Yes	Prokaryotic Harmful bacteria is a result of toxins/chemicals they produce damaging our cells Reproduce asexually via binary fission – Can grow/reproduce inside of cells (intra-) or in tissue fluid, outside of cells (extra-) Classified based on shape Microscopic	No membrane bound organelle Unicellular	Tuberculosis Whooping cough Tetanus
Fungi	Yes	Eukaryotes Generally microscopic if pathogenic	Have a cell wall – chitin not cellulose No chloroplast/chlorophyll, so break down organic matter in other organisms to survive	Athlete's Foot Ringworm Blastomycosis
Protozoa	Yes	Eukaryotes Forms include flagellum or cilia Generally reproduce via binary fission Microscopic	Unicellular	Malaria Toxoplasmosis Amoebiasis
Parasite	Yes	An organism that lives in or on an organism and benefits by deriving nutrients at the host's expense Macro and microscopic Endoparasites – living inside host's body Ectoparasites – living outside of the host's body	Variety of shapes eg. worm-like, bug-like	Scabies Tapeworm Lice

Modes of Transmission

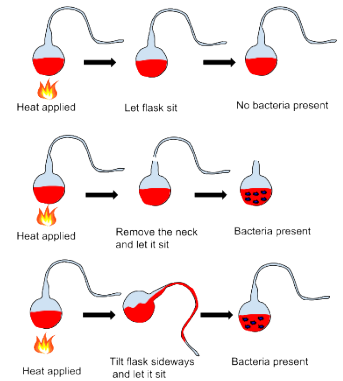
- **Agent** – type of pathogen
- **Reservoir** – area where pathogen resides eg. people, equipment, water
- **Exit** – how it exits an infected person, eg. excretion, secretion, droplets
- **Transmit** – direct (ie. contact, droplet, sexual contact, direct contact with bodily fluids) or indirect (ie. airborne, vector borne, fomite, contaminated food/water)
- **Entry** – eg. respiratory tract, GI tract, mucous membrane, broken skin
- **Host** – what makes someone susceptible, eg. chronic illness, immunocompromised, decreased ability to fight disease for elderly/young people

Entry and Exit:

- **Respiratory** – mouth, nose, throat and lungs; pathogen may lead to irritation, mucus and can exit via coughing and sneezing
- **Gastrointestinal tract (eating)** – stomach, small/large intestine, mouth and rectum as entry and exit points; pathogen may cause vomiting or diarrhoea
- **Genitourinary (peeing)** – reproductive and urinary systems; pathogens enter via urethra or vagina and spread through sexual contact or shared bodily fluids
- **Skin** – if broken entry or exit point; pathogens can spread throughout body

Louis Pasteur, 1822-1895

- Founder of germ theory, father of microbiology
- **Germ Theory** = microorganisms can cause disease, as well as make food spoil
 - Discredited/disputed the theory of spontaneous generation – the idea that things appeared from no where, eg. maggots in bread
- Swan neck experiment – showed that sterilised broth would stay microbe-free if not exposed to the air
 - When it was exposed, the broth spoiled (microbial growth)
 - Thus, demonstrated that microbial germs are airborne and life cannot be generated from a sterilised medium
- Created vaccines for anthrax, rabies and cholera

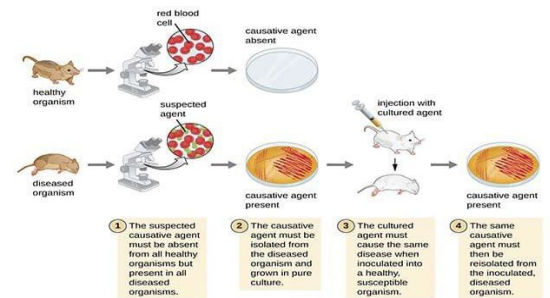


Robert Koch

- Determined causative nature of disease by providing the first evidence that bacteria causes diseases in animals
- Isolated bacteria and cultured them, to which he put in a healthy animal, which then got the disease – proved microbes caused disease
- Worked on anthrax, discovering its rod shape
- His experiments established relationships between particular pathogens and specific disease

Koch's Postulates

- Koch developed principles of disease and summarised these in his four postulates
 - Koch's postulates highlight that transmission of an infectious disease occurs when an infected individual passes on the causative pathogen to a non-infected host
1. The microorganism must be found in all organisms suffering from the disease, but should not be found in healthy organisms
 - a. ie. if all infected organisms don't have the same bacteria, that bacteria can't be the causative of the disease
 2. The microorganism must be isolated and identified from a diseased organism and grown in pure culture
 3. Samples of the organism taken from pure cultures must cause the same disease (displaying the same symptoms) when inoculated into a healthy animal in the lab
 4. The microorganism must be isolated from the inoculated animal and must be identified as the same microorganism that was first identified in the originally diseased host



Impact of Disease on Agricultural Production

Causes	Effect
- Loss of genetic biodiversity, due to monoculture, results in a lack of resilience to diseases - Intensive agriculture practices such as high density livestock facilitates the easy transmission of diseases (close contact) - Increased globalisation results in mobility of diseases through people carrying infected plants and food products across borders (lack of biosecurity) - Wide use of pesticides / antibiotics can result in resistant strains of diseases, increasing their prevalence and severity	- Death and illness in plants and animals - Decreased yield and quality of crops - Loss of trading opportunities both domestically and internationally resulting in a downturn in economic activity and financial hardship for farmers - Food security can be impacted e.g. potato famine - Rise in prices for fresh produce = fall in the health of the population

Plant Disease – Potato Blight of the 1800s

- 1800s – The Irish imported a variety of potatoes from South America, as potatoes formed the basis of the working population's diet
- The potatoes were grown asexually via vegetative propagation, meaning all potatoes were genetically identical
- The potatoes were infected by the fungal disease *Phytophthora infestans* which spread through spores and infected tubers, and destroy $\frac{3}{4}$ of the crop due to the lack of GV
- 1 million Irish people between 1845-1851 died via mass starvation and the poisoning of 100,000 people, and millions more were impacted by a lack of food security

Animal Disease – Foot rot in Sheep

- Caused by the bacteria *Dichelobacter nodosus* which is spread from foot to foot via damp pasture or mud that provide the ideal environment for the bacteria to thrive
- This bacteria causes painful abscesses between the toes, leading to weight loss due to the inability to graze
- Lead to decreased agricultural yields: decreased mutton/lamb quality of infected sheep, reduced wool quality and growth, poor ewe fertility, poor growth rates and reduced value of sale sheep -> decreased economic yields and trade opportunities

Fungal Disease – Dry bubble disease in mushrooms

- Fungal disease that affects mushrooms, spread via sticky spores that are spread quickly via wind-blown dust, water splashes and mushroom to mushroom contact
- Causes deformed and undifferentiated masses of mushroom tissue, making them inedible, decreasing quality and amount of mushroom crops -> decreased production efficiency and value -> decreased economic yield

Responses to Pathogens

IQ 2: How does a plant or animal respond to infection?

- investigate the response of a named Australian plant to a named pathogen through practical and/or secondary-sourced investigation, for example:
 - fungal pathogens
 - viral pathogens
- analyse responses to the presence of pathogens by assessing the physical and chemical changes that occur in the host animals cells and tissues

Jarrah (*Eucalyptus magmata*)'s response to Root Rot (Fungi - *Phytophthora cinnamomi*)

- Root rot releases zoospores which infect plant roots by entering the root bind the rod tip
 - Require water so more likely in moist roots, the environment that the Jarrah tree provides (high moisture content in roots is an adaption to the arid landscape of their native home)
- Symptoms – pale green leaves that wilt and fall readily, shoots die back from the tips, black feeder roots, loss of vigour

Plant Defences of Jarrah

Passive

- Physical Barriers - thick cuticles, small stomata, cell walls, thick bark layers
- Chemical – enzymes to break down toxins, chemical receptors can identify some pathogens which lead to active response

Active

- Recognition via fragments of fungal/bacterial infections
- Rapid – release of H₂O₂ which kills nearby pathogens, cells reinforce their cell walls with nearby materials, cells may surround the pathogen and die off (apoptosis) isolating the pathogen
- Delayed response – sealing wounds and lysosomes released (antimicrobial reaction), salicylic acid (memory agent to halt future infections)

Identifying Foreign Material

- 'Pathogen' = an agent capable of disease
- 1st/2nd line of defence do not determine if a foreign entity is a pathogen
 - Pathogens are grouped into the category of "foreign particles/objects"
- Pathogens are recognised by the 3rd line of defence due to their specific antigens

First Line of Defence – External Barriers

- Innate, non-specific barriers preventing entry

Physical:

- Skin
 - Waterproof, dry, seal, penetrated only if broken
 - Keratin matrix – knitted structure that increases strength, making it harder to penetrate
 - Shedding dead skin cells
 - Epithelial lining has tight cell junctions which blocks pathogens entering
- Cilia
 - Tiny hair-like structures protruding from cells
 - Beat in swaying rhythm pushing mucus and trapped pathogens upwards to be removed easily via sneezing/coughing
 - Trap foreign material
- Peristalsis
 - Muscular motions, rhythmically moving along the oesophagus assists the production of mucus
 - Muscular contractions in gastrointestinal tract
 - Stop microbes from colonising
- Mucous Membranes – lines various tracts
 - Allows substances to enter if necessary
 - Trapped pathogens can be sneezed/coughed out
 - Mucous itself can inhibit growth of harmful pathogens
 - Can contain enzymes

Chemical:

- Sebum (Sweat: antibacterial) and Cerumen (Ear wax and oil on skin)
 - Cerumen – pH of 4, acidic kills microbes
 - Sebum (sweat) has protein dermacin which binds bacteria
- Lacrimal secretions (tears)
 - Contains enzymes which inactivates microbes
- Acidic Secretions
 - Stomach acid: pH 2: kills microbes
 - Duodenum: pH 8.5: kills microbes that thrive in acidic conditions
- Lysosomes (in tears, saliva, mucus)
 - Act on breaking down foreign matter
 - Destroy cell wall of bacteria
- Urine
 - Sterile, slightly acidic, flushes and cleans urinary tract to prevent micro growth

Second Line of Defence – Inflammation

- Begins when first barrier becomes compromised and the barrier is breached
- Non-specific, innate
- Purpose: Swelling and close blood vessels isolate pathogens to site, therefore limiting spread, especially preventing pathogen from entering from bloodstream

Inflammation

- Initiated by infected cells releasing chemicals histamines and prostaglandins
 - Alter the permeability of our blood vessels to increase transport of proteins and white blood cell to the site
 - Act on blood vessel inducing vasodilation (dilation of blood vessel) resulting in higher level of blood flow through the site of infection
 - Allows the body to attack the pathogens as the phagocytes destroy the infection

1. Wound – 1st barrier breached
 2. Cytokines, platelets, mast cells signal the path to injury site
 3. Phagocytes leave bloodstream and engulf pathogens
- As blood is hot, this increased blood flow increases temp. of environment surrounding the infected cells at the infected site
 - Slows the rate at which pathogens produce their enzymes/proteins
 - Histamines also increase the permeability of capillaries which allows more WBC to move to the infected cells

Phagocytosis

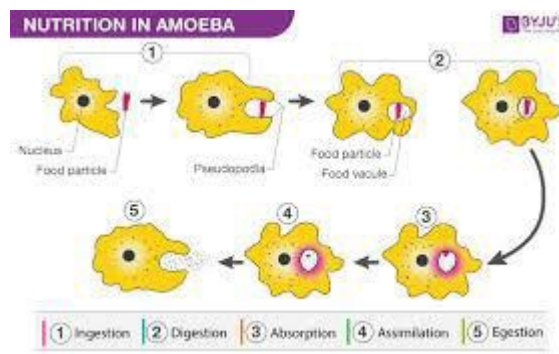
- Carried out by phagocytes (specialised WBC) – neutrophils and macrophages
 - Neutrophils: Short term, move to site of infection, self-destruct after engulfing pathogens or a few days if not used, useful for acute infections
 - Macrophage: longer lasting, used in fighting chronic infection, engulf and destroy pathogens and then display indicators of infection for 3rd line of defence
- Then bind with lysosomes which contain digestive enzymes (eg. Protease) that breakdown antigen
- Advantage – B/T cells take a while to initiate their binding process, especially if it is a primary exposure, so process of phagocytosis is good at minimizing the quantity and damage caused by pathogen before adaptive immunity is activated

Cell Death (Apoptosis)

- Involves neighbouring cells forming a wall of dead cells around infected cells, forming a capsule known as granuloma, trapping the infected cells inside
 - Deprives pathogen of a food source leading to death

Lymphatic System

- The role of the lymph system is to filter and return intercellular fluid to blood using lymph nodes connected by lymph vessels
 - Lymph nodes – where B and T lymphocytes are stored
- Filters and traps antigens



Immunity

IQ 3: How does the human immune system respond to exposure to a pathogen?

- investigate and model the innate and adaptive immune systems in the human body
- explain how the immune system responds after primary exposure to a pathogen, including innate and acquired immunity

The Third Line

- Activates when pathogens breach 1st line and when 2nd line signals for it
- Specific line of defence which is adaptive, which then remembers each pathogen it fights to be better prepared for any possible future infections (specific memory)

Innate VS Adaptive

- Innate:
 - Provides a nonspecific and congenital line of defence against pathogens
 - Genetically programmed
 - With repeated exposure, the body responds the same way

- Adaptive:
 - Cells of adaptive immune system have immunological memories, they remember a pathogen from its previous attack and builds up a response against it the next time specific pathogen enters the body
 - Humoral Response:
 - In fluid, effective against pathogens in body fluids (humors)
 - Cytotoxic Response:
 - In cell, effective against intracellular pathogens
 - Consists mostly of B & T lymphocytes (specialised WBC)

Adaptive Immune Response

- Once the pathogens, enter the 'host', the body is able to determine to the pathogen to be a foreign organism, due to the antigen (protein markers on pathogen, illustrate that it is foreign and that stimulate response)
- When a cell is infected, chemical indicators
 - Interferons – signalling proteins released in response to a pathogen invasion
 - Chemokines - chemical proteins that are released to warn uninfected nearby cells, control messaging for 2nd and 3rd line defence
 - MHC (major histocompatibility complex) I&II – specialised markers on cells so they can present the antigen

B Cells – Humoral Response

- Develop and mature in bone marrow, and also matures in lymph nodes, circulates in the blood
- Responsible for the adaptive immune response between cells

B lymphocyte

- Activated by Helper T cells
- Stimulates B cells to divide and differentiate into either B plasma or B memory

B plasma Cells

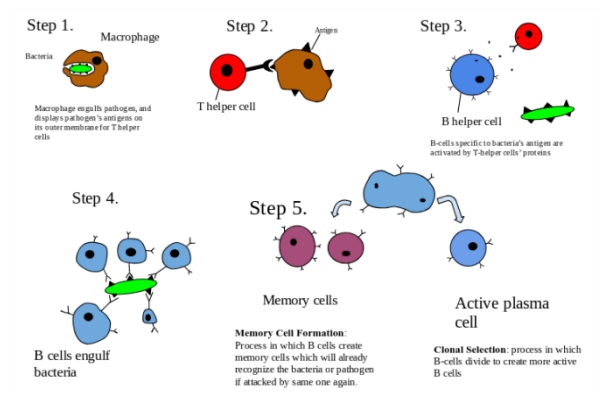
- Provide immunity
- Produce massive amounts of specific antibodies that will bind to their specific antigen

Memory B cells

- Live a long time in body
- Secrete specific antibodies when activated

3rd Line – Primary Immune Response & B-Cells

- B-Cells inactivate pathogens when antibodies on its surface bind to a specific foreign antigen
- B-Cells then become activated with the help of Helper T Cells which releases cytokines
- B-cells then makes many copies of itself which differentiate into:
 - Plasma B cells: produces antibodies (immunoglobulin) against the pathogen
 - Memory B cells: remain in body for long periods of time, provide immunity for subsequent infections (eg. 2nd exposure), storage system
 - Vaccines skip to this stage

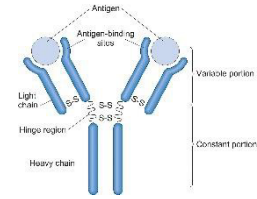


3rd Line – Secondary Immune Response & B-Cells

- When adaptive immune system is exposed to the same pathogen on subsequent occasions, there is a more rapid response and a greater production of antibodies

Antibodies

- Not all antibodies produced are the same and not all antibodies are found in equal concentrations
- Y-shaped protein molecules that consist of 4 chains of protein
 - 2 larger heavy chains
 - 2 smaller light chains
- An antibody molecule has two binding sites, each site is specific for a particular antigen
 - Pathogens have different cell markers on their surface that will bind to different antibodies
 - When bound (coagulated), the resulting molecule is called the antigen-antibody complex
- 5 types of heavy chains (IgA, IgE, IgD, IgG and IgM)
- Different classes act in different ways to neutralise a pathogen:
 1. Neutralisation – antibodies bind to and coat pathogens, blocking their activity
 2. Agglutination – neutralised pathogens clump together and are surrounded by antibodies
 3. Precipitation of dissolved antigens
 4. Activation of the complement system, leading to cell lysis
 5. Opsonisation- enhanced phagocytosis by natural killer cells (NK)



T-Cell

- Develop in bone marrow, mature in the thymine

Killer (Cytotoxic) T Cells

- Kill body cells infected with pathogen/damaged cells
- Releases chemicals (perforin) that destroy foreign, infected and altered cells by making holes in the cell membrane and causing cell lysis

Helper T Cells

- Stimulated by antigen-presenting cells
- Communicate with other immune cells and produce chemicals (interleukin) to stimulate phagocytes, B cells and cytotoxic T cells

Memory T Cells

- Remain in the body after a primary exposure to an antigen to allow a more effective secondary response should re-exposure to the same antigen occur
- Provide long-term immunity

Regulatory T Cells (Suppressor T Cells)

- Immunosuppressors that help control/turn off the activity of immune cells at the end of an immune response

Cell Mediated Immunity (cellular invasion and T-cells)

- Responsible for elimination of pathogens within host cells, regulated by T cells (lymphocytes), mediated by T cell receptors (TCR)
- T cells recognises fragments of/partially digested antigens only
- T cells make direct contact with antigen fragments via TCR
- Receptor binding triggers signal transduction (activity of other immune system cells/blood cells cascading effect of how response will happen) in the T lymphocyte, resulting in proliferation (rapid increase of T cell), cytokine release and activation cytotoxic function
- Healthy cells have Major Histocompatibility complex (MHC1) protein on their surface
 - When cells are infected/cancerous they don't produce that protein
- MHCII is found in antigen-presenting cells (eg. macrophages, dendritic cells and activated B cells)

Cell Mediated Immunity

1. **Engulf:** foreign material engulfed by macrophages, antigens attached to their MHCII molecules
2. **Inspect:** Antigen-presenting macrophages move to lymph nodes, and are inspected by helper T-cells that have the T cell receptor corresponding to the antigen being presented

3. **Activate:** Helper T-cells activate cloning of cytotoxic T cells and memory T cells that are specific for this antigen
4. **Attach:** Cytotoxic cells leave lymph nodes and migrate to site of infection
 - a. Antigen receptors on Cytotoxic cell bond with antigens on infected cell
5. **Destroy:** T cells then release chemicals that destroy the cell and pathogens within it
6. **Clean up:** These chemicals also increase inflammation and attract more macrophages to carry out phagocytosis (clean up of dead cells)
7. **Protect:** Some cytotoxic T cells produce a chemical called interferon, which protects the healthy cells around an infected cell from viral invasion
8. **Calm:** Once infection defeated suppressor T cells release other chemicals to stop the production of cytotoxic T cells
9. **Remember:** Memory T cells specific to that particular antigen remain in the body (lymph nodes)
 - a. On re-exposure to the same antigen-containing pathogen, they cause rapid production of more of the same cytotoxic cells
 - i. Prevents body from developing symptoms of the disease again

Basic Summary of all Components

- Histamines allow for permeability, allowing things like macrophages and neutrophils to exit bloodstream
- Macrophages engulf and trap invaders in membrane, using lysosomes to dissolve
- Neutrophils trap and kill cells
- Dendritic cells present pieces of antigens to signal
- Helper T Cells bind to dendritic, activate and duplicate,
 - Some making memory
 - Some go to fight infection as well as activating B cells
- B cells use antibodies to bind to surface of specific pathogen
- Antibodies flood site, macrophages clean up and fight
- Once infection is defeated, memory cells get stored to have a more rapid response to second exposure

The Lymphatic System

- Arrangement of interconnected tissues and organs that aid in the removal of toxins, cellular wastes, excess fluids and pathogens
- Lymph nodes filter the lymph and remove unwanted material from the body
- Transports WBC, immune cells, including antigen-presenting cells, throughout the body
- Where antigen recognition by lymphocyte occurs
- Roles:
 - Returning fluid that seeps out of the blood vessels into tissues back to the circulatory system
 - Absorbing and transporting fatty acids
 - Place for lymphocytes to mature
 - Transporting lymphocytes and antigen-presenting cells to the lymph nodes stimulating the adaptive immune response

Primary VS Secondary Exposure

- **Primary:** when body 1st experiences a pathogen, the primary immune response creates cytotoxic T-cells and plasma B-cells (producing antibodies)
 - Production of antibodies is relatively slow and low in numbers
 - After infection, left with multitude of T/B lymphocytes
- **Secondary:** memory cells are able to detect the pathogen early and the response by the immune system is far more effective
 - Production of antibodies is much quicker and far greater in numbers
 - Response kills any pathogens well before the pathogen can take hold and begin to harm/damage the host (before symptoms occur)

Vaccines

- Create an artificial experience of primary exposure, synthetic made to mimic a natural response
- Once body has overcome the vaccine, the memory cells that remain in the body to more effectively fight off any future infections
- 1. Live attenuated vaccines
 - a. Use living but weakened pathogens (mumps, chicken pox, measles)
 - i. Must be stored well as living
 - b. Most like natural infection and provide a strong disease immunity
- 2. Inactivated vaccines (eg. Hep A, flu, polio)
 - a. Dead or killed pathogens
 - b. May need several doses/booster shots over time
- 3. Toxoid vaccines (eg. diphtheria, tetanus)
 - a. Protect against harmful substances (toxins) produced by pathogens
 - b. Use weakened versions of the toxins (toxoids)
 - c. May need booster shots
- 4. Subunit and conjugate vaccines (pertussis part of DTaP vaccine and the hepatitis B vaccine)
 - a. Specific part of a pathogen
 - b. Provide strong immunity to that key part of the germ
 - c. May require booster
- 5. mRNA vaccines (eg. covid-19 vaccine like Pfizer and Moderna)
 - a. Contain mRNA instead of using germs or toxins
 - b. mRNA instructs your cells to make the germ's antigen which triggers an immune response
 - c. Does not carry live virus and do not change your DNA
- 6. Viral vector vaccines (eg. Astra Zeneca)
 - a. Use DNA or mRNA to instruct cells to make germ's antigen
 - b. Genetic material wrapped in a different safe virus to take instructions into your cell
 - c. Does not change DNA

Complement System (Complement Cascade)

- Made up of 30+ proteins, part of immune system that helps (complements) the ability of antibodies and phagocytic cells to trigger:
 1. Membrane attack - by rupturing cell wall of bacteria
 2. Phagocytosis – by opsonizing antigens
 3. Inflammation – by attracting macrophages
- Not adaptable, non-specific and present at birth
- Can be recruited and brought into action by antibodies

Cytokines

- Proteins produced by immune cells (eg. macrophages, neutrophils, B cells, T cells)
- Act as chemical messengers in innate and acquired immune system
- Roles:
 - Attract other phagocytes to area of infection
 - Stimulate activity of natural killer cells
 - Inhibit activity of WBC until infection is defeated
 - Interferon cytokines interfere with the reproduction of viruses

Major Histocompatibility Complex (MHC) Molecules

- MHC 1 – signal ‘self’ cell
 - Found on all nucleated cells that present normal self-antigens as well as abnormal/non-self pathogens to the effector T-cells involved in cellular immunity
 - Healthy cells – proteins found in cytoplasm are degraded by proteasomes and processed into self-antigen epitopes, which bind within the MHCI antigen-binding cleft to be presented on cell surface
 - Immune cells like NK cells then recognise these self-antigens and not target cells for destruction
 - If the cell becomes infected with an intracellular pathogen, protein antigens specific to the pathogen are processed in the proteasomes and bind with the MHCI molecules for presentation on the cell surface
- MHC II
 - Only found on macrophages, dendritic cells and B cells
 - Present abnormal/non-self pathogen antigens for the initial activation of T cells

Prevention, Treatment and Control

IQ 4 – How can the spread of infectious diseases be controlled?

- investigate and analyse the wide range of interrelated factors involved in limiting local, regional and global spread of a named infectious disease
- investigate procedures that can be employed to prevent the spread of disease, including but not limited to:
 - hygiene practices
 - quarantine
 - vaccination, including passive and active immunity
 - public health campaigns
 - use of pesticides
 - genetic engineering
- investigate and assess the effectiveness of pharmaceuticals as treatment strategies for the control of infectious disease, for example:
 - antivirals
 - antibiotics
- investigate and evaluate environmental management and quarantine methods used to control an epidemic or pandemic
- interpret data relating to the incidence and prevalence of infectious disease in populations, for example:
 - mobility of individuals and the portion that are immune or immunised
 - Malaria or Dengue Fever in South East Asia
- evaluate historical, culturally diverse and current strategies to predict and control the spread of disease
- investigate the contemporary application of Aboriginal protocols in the development of particular medicines and biological materials in Australia and how recognition and protection of Indigenous cultural and intellectual property is important, for example:
 - bush medicine
 - smoke bush in Western Australia

Key Definitions

- Prevention – Efforts/strategies to stop the occurrence/onset of a disease
- Treatment – Administration of remedies (pharmaceuticals, behavioural, surgical, etc) to aid an individual suffering from a disease
- Control – Contain the spread of a disease in a given population
- Epidemic – Localised within a region, over a particular season or time period
- Pandemic – Large scale outbreak (eg. across continents) declared by WHO
- Incidence – The number of new cases occurring within specific population within a specific time
- Prevalence – the proportion of the population that has the disease at a particular time (both previous and current – all cases)
- Zoonotic disease – disease passes from animal to human

Factors impacting Emergence and Spread of Diseases

- Human Migration
 - Movement of people due to war or socio-economic reasons facilitates the spread of pathogens to new, uninfected areas
 - Pathogen then exposed to new hosts, who might not have the same immunity
 - As globalisation has increased and more people travelling across larger geographical areas in short periods of time, emergent diseases have the potential to spread more rapidly
 - Rapid movement also makes it harder to trace the origins for an emergent disease
- Human Behaviour
 - Impacts emergence and reemergence of disease and increases the spread
 - Includes:
 - Hygiene practices, sanitation, dietary habits, human-human contact, sexual activity, medical procedures, exposure to environmental agents of disease
 - Also increasing rates of urbanisation as humans continue to move to urban areas, increases the density of populations, facilitating disease to spread more rapidly
- Farming Practices and Food Productions
 - Close proximity to human populations allows diseases in farm animals to transfer to humans more easily
 - New zoonotic diseases have severe effects as humans have no previous resistance
 - Highlights a need for worldwide surveillance of animal and human viruses on local, regional and global scales

Prevention

- Hygiene – slow rate of spreading contagious diseases and preventing disease spread
 - Personal –
 - Washing hands with disinfectants to eliminate pathogens
 - Covering mouth and nose when sneezing/coughing to prevent spread
 - Not meddling with open/healing wounds
 - Not sharing cutlery/napkins/tissues etc
 - Preparing/cooking fruits/vegetables/meats properly
 - Disposing garbage properly
 - Community –
 - Garbage/Sewerage disposal facilities
 - Sterilisation/Disinfectant of equipment including at hair dressers, dentists and hospitals
 - City planning to reduce overcrowding
 - Strict guidelines for food handling, storage and serving
- Quarantine – limit spread by preventing entry of risky items or isolating infected hosts until pathogen is eliminated, thus preventing transmission from host to host
- Public Health Campaigns – raising awareness and implementing public health programs
 - Government regulations ensure standardized procedures are followed, eg. food handling, sterilizing of equipment for treatment
 - Reporting of disease to allow for strategies to be put in place
 - Public immunization programs
 - Screening of diseases, maternal and child health programmes, family planning
- Use of pesticides (agriculture)
 - Kill pathogens or eliminate vectors that transmit them
 - Cons: ability of vectors/disease causing organisms to build up resistance, environmental damage
- Genetic Engineering (agriculture)
 - Altering genetic composition of an organisms to make them disease resistant
 - Eg. transgenic species

- Vaccination
 - To gain acquired immunity and prepare host for future challenges from a pathogen
 - Each vaccine is specific for one type of antigen
 - Herd immunity can be achieved
 - The indirect protection from an infectious disease that happens when a population is immune through vaccination

Antibiotics

- Treat diseases caused by bacteria
- Non-specific: synthetic drug that attaches and attacks anything with a cell wall, including 'good' bacteria
- Work by blocking replication by preventing cell wall, nucleic acid or protein synthesis in bacteria, killing the bacteria or stopping them multiplying → helps the body's natural immune system to fight the bacterial infection
- Effectiveness:
 - Target pathogenic bacterial cells rather than human cells, thus have very limited side-effects
 - Can be used to prevent secondary infections or to avoid infections in surgical infections

Antibiotic Resistance

- Occurs when bacteria mutate and become less sensitive to an antibiotic
- The more antibiotics used/taken incorrectly = more likely it is that resistant bacteria will survive, grow and multiply, making future bacterial infections harder to treat
- How to reduce:
 - Take antibiotics only for bacterial infections
 - Take antibiotics as directed by health provider
 - Take the right antibiotic

Antivirals

- Used to treat viral diseases by blocking action during infection, including:
 - Entry of virus to cell [strengthens cells' ability to not be invaded]
 - Reverse transcription of viral RNA/DNA
 - Translation of viral DNA into proteins
 - Integration of viral DNA into the genome of cell
- Some are virus specific, some are broad-spectrum
- Effectiveness:
 - Cannot cure disease but can decrease duration and reduce symptoms by assisting immune system to fight against the virus
 - As a preventive measure, it can decrease likelihood of sickness by 55% and can reduce risk of developing severe illnesses and reduce chances of that specific viral turning lethal for that patient

Ebola Outbreak 2014-2016 in West Africa

- Ebola virus (ssRNA virus)
- Highly contagious, causes rapid death but is easily preventable
- Spreads via direct contact with body fluids and mucous membranes, including sexual activity
- Believed to be a zoonotic disease, with transmission mostly like due to close human contact with wild animals who acted as reservoirs of the virus
- Symptoms – fever, tiredness, sore throat, which progresses to vomiting, diarrhoea and oozing of blood from mucous membrane/stools

- Prevention:
 - Good sanitation and hygiene
 - When Ebola reached the USA, it did not cause an epidemic as it did in West Africa, as America as they had the education required to know about modes of transmission and nature of the virus, and the public infrastructure to tackle cases and reduce spread
 - In West Africa, as well as having poor public health infrastructure and low levels of education on the nature/spread of the disease, cultural practices such as washing the dead, also provided a means of transmission
 - Also, may prevent individuals from seeking care at hospitals, instead favouring traditional and spiritual healers
- Control:
 - Administrative – initial organisation of response and obtaining funding
 - Environmental – provision of facilities for barrier nursing (providing care in strict infection control conditions), water and hygiene control, hand hygiene and safe waste management
 - Provision of the correct PPE
 - International aid -set up Ebola clinics and supply PPE, medical resources, helped isolate and aid treatment of cases
 - Quarantine – isolating infected persons, assigning the same clinical staff to each patient, visitor restriction
 - Longer term – reducing wildlife-human transmission by ensuring meat is cooked thoroughly and wearing PPE when handling reservoirs

Malaria

- Disease caused by protozoan of Plasmodium genus, spread via vector of female mosquito (which thrive in tropical environments)
- Causes fevers, chills, diarrhea, confusion and seizures
- Managed through medicine
 - Artemisinin-based combination therapy (ACT): Combining 2+ drugs that work against malaria in different ways
- Prevented:
 - Reduce contact between humans and mosquitos (nets, PPE, insect repellent, destruction of larvae)

Dengue Fever

- Viral disease spread via vector mosquito, common in tropical and subtropical areas
- Develops into a fever with flu-like symptoms

Historical, Cultural And Current Strategies to Predict/Control Spread of Disease

- Historical:
 - Lack of scientific knowledge/technology meant decisions/theories were made without adequate understanding
 - Miasma (Bad air) Theory which is now discredited due to germ theory
 - Cultural/religious values held more standing than scientific principle, eg bubonic plague (500-1450AD) attributed to 'God's wrath' rather than pathogen
 - 14th C – a link between plague and movement of symptomatic people led to first quarantine practices with ships
 - 1854 John Snow – show a common pattern for people with cholera through spot mapping and documenting cases
 - Linked proximity of cases to people using water pump on Golden Square, London
 - Predicted water pump was source of cholera and convinced council to close off pump, leading to a significant reduction of cases
 - Lead to discovery that cholera is a water-borne disease

- Cultural (ideas, customs, traditions, behaviours held by a group of people):
 - Boiling water –4000 years ago in Greek, Roman and Chinese cultures
 - Herbs believed to have medicinal benefits eg. Kakadu plum in Indigenous cultures
 - Cultural practices of washing the dead in West Africa led to an increase in Ebola cases
- Current: Increase in scientific knowledge = increase understanding of pathogens and modes of transmission = better informed procedures/policies to decrease spread so epidemics can be controlled
 - Eg Dengue fever/Malaria has been controlled better due to understanding of climate and mosquitos and pathogens
 - WHO governs widespread pandemics and notifiable diseases
 - R_0 value:
 - <1 – decline in disease (will die out) as existing infectious person will cause less than one new infection
 - $=1$ – disease ‘stable’ as 1 infected person will infect 1 person
 - >1 – spread is increased as one person gives disease to more than one person

Indigenous Treatment Methods

- Bush medicine –
 - Made from herbal extracts and used in the maintenance of health and as a preventive/treatment measure for disease
 - Both ritual and remedy, apart of Indigenous culture and encompasses natural environment and spiritual realm
 - Used still to treat basic diseases and as first aid
 - Traditions largely lost in southern and eastern Aus, but traditional practices are being documented by anthropologists in western Aus
 - Serve as raw materials for some drugs that are developed using modern pharmaceutical techniques (crude drugs)
 - Eg. Tea Tree – antiseptic oil, Kakadu plum – rich in Vit E
 - Regarded as intellectual property but not currently protected by adequate legislation and therefore can be exploited by commercialism
- Smoke Bush (*Conospermum sp*) –
 - Native shrubs grown in coastal Western Australia known for its healing properties
 - Contains Conocuvone which has the potential to destroy low conc. of HIV
 - Private pharmaceutical company has been given a global license to patent a product from Conocuvone
 - When commercially successful, both private company and Western Aus gov will benefit from profits and royalties

Module 8: Infectious Diseases

Homeostasis

IQ 1: How is an organism's internal environment maintained in response to a changing external environment?

- construct and interpret negative feedback loops that show homeostasis by using a range of sources, including but not limited to:
 - temperature
 - glucose
- investigate the various mechanisms used by organisms to maintain their internal environment within tolerance limits, including:
 - trends and patterns in behavioural, structural and physiological adaptations in endotherms that assist in maintaining homeostasis
 - internal coordination systems that allow homeostasis to be maintained, including hormones and neural pathways
 - mechanisms in plants that allow water balance to be maintained

Homeostasis

- Process by which organisms maintain a relatively stable internal environment
- Two Key Steps:
 1. Detecting the change: the stimulus detected using receptors
 2. Counteracting the change: will req. an appropriate responses, in order to reverse/counteract the change and return to normal (optimum) range
- Consists of 5 minor stages:
 1. Stimulus: the change in environment (int. or ext.). eg. change in temp
 2. Receptor: specialised organs/cells within the body, used to detect any variation for the 'normal' (optimal cycle)
 3. Control Centre: usually within the brain (the hypothalamus), this interprets the changes/determines the most effective response to 'reverse' the change
 4. Effector: component of the body that reacts (eg. muscle or gland) and cause the response
 5. Response: carried out by the effects, in order to return the body to correct temp or sugar level
 - a. Can be a cascade response

Negative Feedback Loop

- Mechanism ins body that help maintain homeostasis, by reversing changes in body back to normal range (threshold, optimum)

Temperature and Homeostasis

	Hyperthermia	Hypothermia
Stimulus	Increase in body temp	Decrease in body temp
Receptor	Thermoreceptors in the brain	Thermoreceptors in the brain
Control Centre	Hypothalamus activates cooling mechanism	Hypothalamus activates heating mechanism
Effector	Sweat glands Vasodilation Decreased metabolic activity	Vasoconstriction (blood vessels away from surface) Increased metabolic activity Shivering (fine motor skill movements to generate heat)
Response	Decreased body temp to return to normal range and maintain homeostasis	Increased body temp to return to normal range and maintain homeostasis

Glucose and Homeostasis

- Importance of glucose:
 - Body's primary fuel source
 - Constant maintenance of glucose levels ensures optimal bodily metabolic function
 - Reactant in aerobic respiration
- Too high – lead to heart attack, stroke, nerve/eye/kidney damage
- Too low – weakness, light headedness, dizziness, seizure and coma

	Hyperglycaemia	Hypoglycaemia
Stimulus	Increase in blood sugar concentration	Decrease in blood sugar concentration
Receptor	Beta pancreas cells	Alpha pancreas cells
Control Centre	Glycoregulatory neurons in hypothalamus	Glycoregulatory neurons in hypothalamus
Effector	Pancreases beta cells Release insulin and body/liver takes up more glucose	Releases glucagon and liver breaks down glycogen to release glucose in blood
Response	Decreased blood glucose levels, back to set point	Increased blood glucose levels, back to set point

Adaptations

- A specific characteristic/feature/trait that aids in the survival and reproductive success of an organism and species

Endotherm VS Ectotherm

- Endotherm: Organisms that uses energy to regulate its internal body temp. at a relatively constant level regardless of ambient temperature
- Ectotherm: An organism whose regulation of body temperature depends on external temperature

	Structural	Behavioural	Physiological
Description	A physical feature of an organism that aids its survival and helps it maintain homeostasis	Conscious action that occurs to aid the survival of organism by maintaining homeostasis Can be instinctive (innate) or taught (learned)	A process that occurs to regulate and maintain homeostasis in an endotherm or aids in survival of an organism
Example	Polar bear – Thick fur with black skin, blubber to insulate body → maintain an internal 37 degrees in Artic environment	Kangaroo licking forearms/paws to cool down → evaporation of water removes heat → returns internal temp to optimum range	In humans: Vasodilation or vasoconstriction Increase breathing when O2 levels are low Sweating to decrease body temperature → maintain homeostasis

Trends and Patterns with Adaptations

Behavioural

- Koalas adopt different body postures in trees, depending on heat of day/environment
- Hot weather – more of the body is in contact with the tree trunk and large branches, which are cooler than the air. This increases heat loss through the tree
 - As temperature rises, body exposure to tree increases (ie. greater SA) when koalas change their posture = decrease in temperature = maintenance of body heat
- Cool weather – Less of the body is in contact with the tree trunk and large branches. This reduces heat loss through the tree
 - As temperatures decrease, less of the body is exposed to the tree as koalas change their posture to decrease their contact with tree to increase body temperature

Structural

- Surface area of an endotherm's body relative to its body volume, affects the amount of heat lost to the surrounding air
- As environments move from cold/artic to hot/arid, hare populations increase in body surface area to volume ratio to increase heat loss to surrounding air
 - As external temperature increases, size of ear, size of leg increases while fur thickness decreases

Physiological

- ADH – hormone that regulates the amount of water in the body, that when released from pituitary gland, prevents the kidneys from making too much urine = water retention
- Spectacled hare wallaby VS Rothchild's rock wallaby
 - Spectacled hair:
 - Live in spinifex bushes during heat of the day, when air is dry/temp over 40 degrees
 - Rock Wallaby:
 - Lives in cooler caves with a temp about 30 degrees and higher humidity
- ADH trend:
 - Hare wallaby uses ADH as a physiological adaptation to reduce urine production, to retain water in the body
 - Rock wallaby instead relies on reduced blood flow to kidneys instead of hormonal control, which combined with its ability to seek out cool rock shelters, allows this species to regulate the internal water levels

Systems to maintain Homeostasis – Hormones (Endocrine System)

- Chemical messengers created by the endocrine glands that are transported around the body via the circulatory system
- Maintained by hypothalamus, which is the gland that determines whether optimal conditions are being maintained within the body

Plant hormones

Hormone	Functions
Auxin	Promotes cell elongation (vertical growth towards light, phototropism) Helps in cell division/growth
Gibberellin	Promotes cell elongation (vertical growth towards light, phototropism) Helps in cell division/growth Promotes seed germination
Absciscic acid	Maintains seed dormancy Promotes abscission Tells stomata to open/close, allowing for gas exchange

Types of hormones

Hormone Group	Solubility	Chemical Structure	Mode of Action	Eg.
Steroid	Fat	Made from cholesterol	Slow onset, long lasting Travel in blood via protein Diffuse across membrane and bind to nucleus	Testosterone Progesterone
Peptide	Water	Small chain of a.a	Fast acting	Insulin
			Travel in blood Attach to receptors on the membrane	Oxytocin
Amine	Fat/Water	Derivatives of amino acid tyrosine	Shares properties with steroid/peptide hormones	Adrenaline

Types of Hormone Signalling

- Autocrine signalling: Hormone released and detected by same cell, eg. T lymphocytes used to stimulate cloning
- Paracrine signalling: signals sent to cells nearby, eg. for sealing wounds, communication between nearby cells and blood clotting
- Endocrine signalling: cells communicate in different parts of the body
 - Eg menstrual cycle, FSH released by pituitary gland (brain) and travels via blood to be picked up by receptors of ovaries

Endocrine Glands in Humans

Gland	Hormone	Function/Response
Pituitary	FSH	Sexual development and controls menstrual cycle
Thymus	Thymopoietin	Fuels production of T cells and helps pituitary glands to release hormone
Kidney	Renin	Control blood pressure, maintains levels of Na ⁺ /K ⁺ in body
Pancreas	Insulin	Allows glucose to enter cells and maintains glucose levels

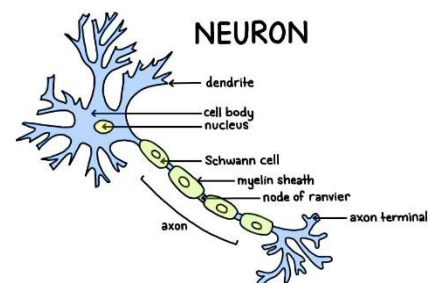
Systems to maintain Homeostasis - Nerves

Nervous System

- CNS – everything involuntary, incl. brain and spinal cord
- PNS – consists of all nerves that are not part of CNS, that branch to organs, glands and muscles
 - Somatic system – voluntary movements (skeletal muscles)
 - Autonomic nervous system – involuntary, ‘flight/fight’ reactions, digestion, heart rate, gastric functions

Neuron

- Dendrites: receive signal from receptor/other neuron
- Axon: sends signal (via action potential) to neighbouring neuron
- Myelin sheath: covers axon and insulates



Types of Receptors

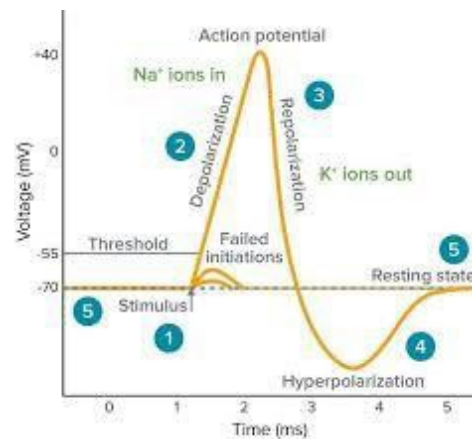
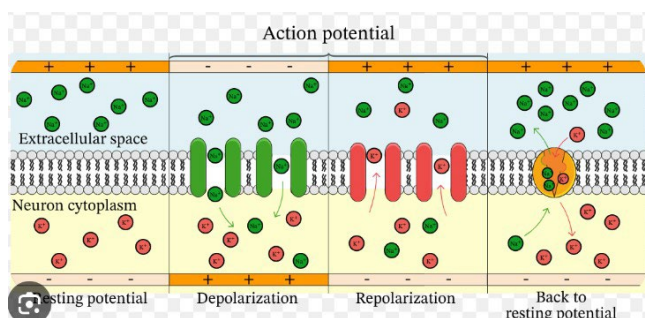
- Interceptors – detect chemical changes inside organs
- Thermoreceptors – detect changes in heat
- Chemoreceptors – chemical changes in blood
- Osmoreceptors – salt, ions and water levels in blood (osmolarity)

How do nerve cells work?

- Functions through an electrochemical impulse messenger
 - When resting, cytoplasm has – charge by surrounding by sodium ions (Na⁺): charge difference = -70mV
- When nerve is triggered (ie stimulus detected), Na⁺ rush inwards
 - Part of nerve cell cytoplasm becomes positive
 - Causes next section to open up to Na⁺
 - Pattern repeats like motion wave
 - As impulse passes the Na⁺ leave the cells and return to resting potential

Action Potential

1. Depolarisation - Sudden change in membrane potential from – to +:
 - In response to a signal initiated at a dendrite, sodium channels open within the membrane of the axon
 - As Na^+ ions are more conc. outside of the neuron, the opening of sodium channels causes a passive influx of sodium
 - Influx of sodium causes the membrane potential to become more positive (depolarisation)
2. Repolarisation – restoration of a membrane potential following depolarisation:
 - Following influx of sodium,, potassium channels open within the membrane of the axon
 - As K^+ ions a more conc inside the neuron, opening potassium channels creates a passive efflux of potassium
 - Efflux causes membrane potential to return to a more negative internal differential (repolarisation)
3. Refractory Period- Period of time following a nerve impulse before neuron is able to fire again:
 - In a normal resting state, sodium ions are predominantly outside the neuron and potassium ions are mainly inside (resting potential)
 - Following depolarisation and repolarisation, this ionic distribution is largely reversed
 - Before a neuron can fire again, the resting potential must be restored via the antiport section of the sodium potassium pump



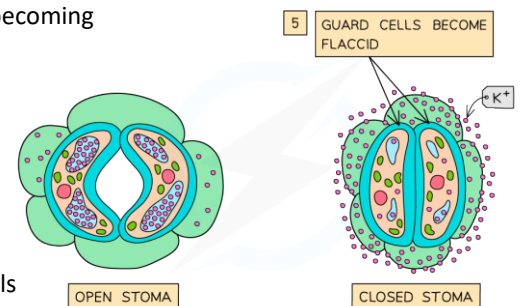
Mechanisms to Maintain Water Balance

- Plant Adaptations:

Leaf surface area	Positioning of stomates	Shape of leaves	Depth/nature of roots
Arrangement of leaves	Thickness of cuticle	Number of stomates	Colour of leaf surface

Regulating Water Loss – Opening and Closing of Stomata

- Guard cells flank the stomata and can occlude the opening by becoming increasingly flaccid in response to cellular signals
- When a plant begins to wilt from water stress, dehydrated mesophyll cells released the plant hormone abscisic acid (ABA)
- ABA triggers the efflux of potassium from guard cells, decreasing water pressure within the cells (lose turgor)
- A loss of turgor makes the stomatal pore close, as the guard cells become flaccid and block the opening



Causes and Effects

IQ 2: Do non-infectious diseases cause more deaths than infectious diseases?

- investigate the causes and effects of non-infectious diseases in humans, including but not limited to:
 - genetic diseases
 - diseases caused by environmental exposure
 - nutritional diseases
 - cancer
- collect and represent data to show the incidence, prevalence and mortality rates of non-infectious diseases, for example:
 - nutritional diseases
 - diseases caused by environmental exposure

Genetic Diseases

- Genetically transmitted and caused by errors in genetic information:
 - Change in chromosome numbers, change to a single gene, single gene abnormality
- Eg.
 - Cystic Fibrosis (see depth study summary)
 - Sickle Cell Anaemia
 - Point mutation within a specific gene, leading to the wrong a.a in the haemoglobin beta protein
 - Leads o changes in blood cells overall structure (crescent, sickle shape) = decreased efficiency in carrying O₂

Environmental Diseases

- Lifestyle, physical factors and chemical exposure
- Lifestyle:
 - Cardiovascular disease caused by substance abuse (eg. alcoholism and smoking) that involves plaque build up that thicken and stiffen the artery walls, inhibiting blood flow to organs
- Physical factors:
 - UV radiation causing skin cancer (see Mod 6 notes)
- Chemical exposure:
 - Lead poisoning – long-term build up of lead in body (eg. due to lead-based paints in older buildings or working in industries like welding)

Nutritional Diseases

- Caused by either undernutrition (insufficient intake of the correct food types) or overnutrition (excessive intake of food)
- Undernutrition:
 - Kwashiorkor: severe form of malnutrition caused by a major deficiency in dietary protein
 - Causes an osmotic imbalance in the lower digestive system, causing swelling of the gut (an oedema) due to a retention of water
 - Iron deficiency: when the body does not absorb enough iron, leads to a lack of oxygen uptake which causes lethargy, fatigue and weakness
- Overnutrition:
 - Overweight/Obesity: abnormal/excess fat accumulation that leads to immobility and chronic health issues

Cancer

- Can be caused by genetics, environment/lifestyle and nutrition
- Arises due to the change of normal body cells into tumorous cells, causing cells to grow and divide uncontrollably
- Changes are the result of a person's genes being impact by different external agents:
 - Physical carcinogens: UV/ionizing radiation
 - Chemical carcinogens: asbestos, tobacco smoke, aflatoxin (food contaminant)
 - Biological carcinogens: a result from certain pathogens eg. HPV

- Genetic background for cancer cells:
 - DNA Repair Genes:
 - Stop the cell cycle and help correct any damaged section of genes
 - If mutated, cells are not able to repair any DNA damage -> impacts functioning of cell and causes issues
 - Proto-Oncogenes:
 - Code for proteins necessary for cell growth/mitosis
 - Mutations within genes lead to uncontrollable cell growth and prevent cell death
 - Tumour-Suppressor Genes:
 - Code for the cell to slow/stop cell growth/mitosis and also code for cell to induce cell death if number of cells become too great
 - Any mutation may lead to continuous cell division
- Risk of cancer can increase from:

Overweight/Obesity	Tobacco use	Unhealthy diet	Alcohol use
Transmissions of HPV or Infection of Hepatitis or infection of other pathogens	Radiation, UV and ionizing	Lack of physical activity	

Epidemiology

IQ 3: Why are epidemiological studies used for?

- analyse patterns of non-infectious diseases in populations, including their incidence and prevalence, including but not limited to:
 - nutritional diseases
 - diseases caused by environmental exposure
- investigate the treatment/management, and possible future directions for further research, of a non-infectious disease using an example from one of the non-infectious diseases categories listed above
- evaluate the method used in an example of an epidemiological study
- evaluate, using examples, the benefits of engaging in an epidemiological study

Type 2 Diabetes – Treatment, Management, possible future directions for further research

- What:
 - occurs when pancreas either cannot make any or enough insulin, and/or insulin it does makes does not work very well, resulting in too much glucose in the bloodstream
- Who's at risk:
 - family history, pre-diabetics, overweight/obese, inactive lifestyle, increases with age, Aboriginal/Torres Strait Islander background
- Symptoms:
 - Excess thirst and urination [extra sugar stored in blood which forces kidneys to work overtime to filter and absorb extra sugar]
- Current management:
 - Healthy eating, regular physical activity
 - Current medication (eg. metformin, pioglitazone) – increase insulin sensitivity and glucose which helps type 2
 - However, some side effects incl. nausea and weight gain
- Future directions of research – Artificial Pancreas
 - New, fully automated insulin delivery system
 - Device is powered by an algorithm and involves a readily available glucose monitor and insulin pump with an app
 - App predicts how much insulin is needed to maintain glucose levels within a target range
 - Study at Cambridge with 26 people found that using the device the overall time spent with blood levels lower than 10mmol/L target doubled
 - First fully automated device with previous systems for type 1 being only semi-automated (hybrid) and eliminates the need for time-consuming manual calculations
 - Simplicity, natural successor to semi-automated

Heart Disease – Future directions for further research

- 1/3 die of heart disease, not enough organ donors
- Previous hearts have working on mimicking the pumping action of the natural heart, which is muscular
 - Short lived as relies on elasticity which is hard to maintain unless connected to nerve
- Daniel Timms' heart mimics a rotor, using magnetism which lets a rotor inside the vein, pumping blood around the body with the parts never touching and never wearing out

Epidemiological studies should:

- Conducted over long periods of time
- Study large samples, collecting a range of data (sex, age, diet, occupation, lifestyle, exercise)
- Have participants that represent the mean population
- Use control groups not exposed to risk/disease
- Data on incidence, prevalence, mortality and morbidity

Analytical Studies

- Cohort study:
 - Prospective
 - Follows a large group of people over a period of time and measures the frequency of the disease in exposed and non exposed individuals (control)
 1. Large group recruited
 2. Each person is interviewed to gather data
 3. Individuals then followed over time, interviews conducted along the way
 4. Data about risks, lifestyles are analysed. A relative risk is calculated
- Case control study:
 - Retrospective
 - Disease status in populations is known and researchers work backwards

Evaluating Epidemiological Studies

Study Type	Benefits/Strengths	Limitations
Case control (retrospective)	Individual satisfaction from contribution to knowledge about the disease for future generations Society benefits from knowledge about risk factors/exposures = increase in scientific knowledge	Don't usually bring direct benefit/improvement to individual Retrospective nature of data collection can lead to bias and relies on recall
Cohort (prospective)	Individual satisfaction Science knowledge increases from data collection and correlating risk factors with development of disease	No individual benefit to individual health Large cohort studies can be expensive, hard to follow up and relies on funding

Errors in Epidemiological Studies

- Random errors – unpredictable variation in data, inconsistent effect on measurement in a study
 - Ensuring groups studied are homogenous will increase precision and reduce random error
- Systematic errors – associated with bias, may be due to sampling bias

Prevention – SECONDARY SOURCES, NO NOTES

IQ 4: How can non-infectious diseases be prevented?

- use secondary sources to evaluate the effectiveness of current disease-prevention methods and develop strategies for the prevention of a non-infectious disease, including but not limited to:
 - educational programs and campaigns

- genetic engineering

Technologies and Disorders

IQ 5: How can technologies be used to assist people who experience disorders?

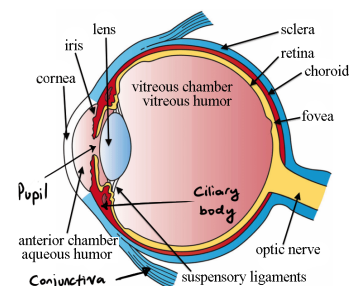
- explain a range of causes of disorders by investigating the structures and functions of the relevant organs, for example:
 - hearing loss
 - visual disorders
 - loss of kidney function
- investigate technologies that are used to assist with the effects of a disorder, including but not limited to:
 - hearing loss: cochlear implants, bone conduction implants, hearing aids
 - visual disorders: spectacles, laser surgery
 - loss of kidney function: dialysis
- evaluate the effectiveness of a technology that is used to manage and assist with the effects of a disorder

THE EYE

Structures and their Functions

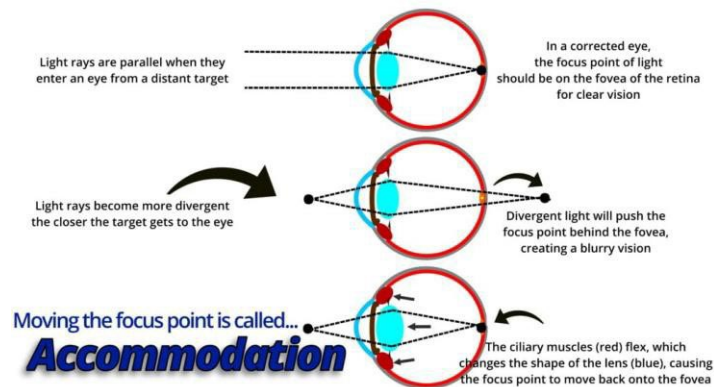
Structure	Description	Function
Cornea	Transparent membrane made from collagen protein Convex lens	Barrier protection Refracts light rays into pupil
Choroid	Thin layer of tissue that is rich in blood vessels	Provides blood supply into the retina
Retina (rods and cones)	Light sensitive layer of tissue at the back of the eye	Detects wavelength of light Converts light to electrochemical signals
Iris	Coloured muscular ring around the pupil	Controls the amount of light that enters eye Dilates (when little light) and constricts (in bright conditions)
Pupil	Opening at the centre of the iris	Allows entry of light into the eye
Lens	Transparent biconvex flexible protein disc shaped	Main refractive structure Focuses light onto the retina
Aqueous and vitreous humour	Transparent water like fluid	Maintains pressure, provides nutrition and refracts light
Ciliary body and sensory ligaments	Ring of tissue containing muscles and ligaments	Hold the lens in position and changes lens' 'thickness' Curvature = allows focusing to occur
Optic Nerve	Paired nerve that connects the eye to the brain	Transmits nerve impulse from retina to the brain, ready for interpretation

- Accommodation [movement of the lens]
 - Refraction occurs in the eyeball where light waves are bent towards the retina to form an image via refractive media
 - Cornea, aqueous humour, lens, vitreous humour
 - Accommodation: Process by which the eye changes refractive power by altering the shape of the lens in order to focus objects at various lengths

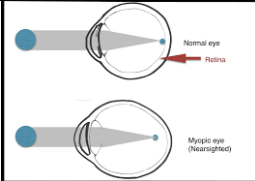
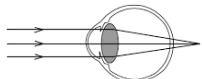
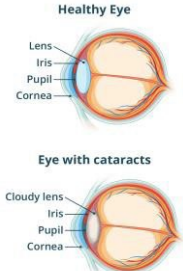
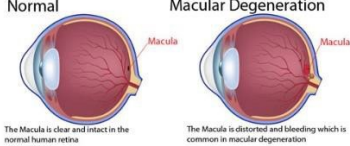


Type of Vision	Shape of Lens	Action of Muscles of Ciliary Body	Tensions of Suspensory Ligament

Distant	Elongated, slightly biconvex	Relaxes	Taut
Near	Highly curved, biconvex lens w high refractive power	Tighten/contracts	Slackened

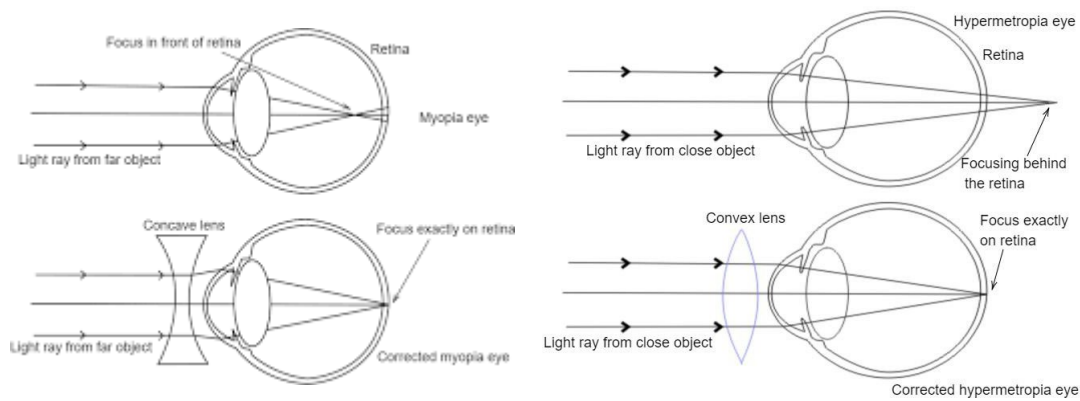


Visual Disorder

Disorder	Description	Cause	Diagram
Myopia	Short sightedness (see objects close to you)	Cornea is too curved Lens is too fat Eyeball shape is too oval ->focal point falls short	
Hyperopia	Far sightedness	Cornea is too thin/not curved enough Lens is not curved enough Eyeball shape is too round	
Cataracts	Blurry vision from breakdown of proteins and fibres in the lens, making it no longer translucent	UV light, smoking, diabetes, age	
Macular degeneration	Deterioration of cells resulting in loss of central vision as photoreceptor cells can't pick up light	Smoking, obesity, high blood pressure, family history	

Technologies

- Spectacles are worn outside the eye so that it can adjust the refraction of light prior to the radiation striking the eye and undergoing refraction

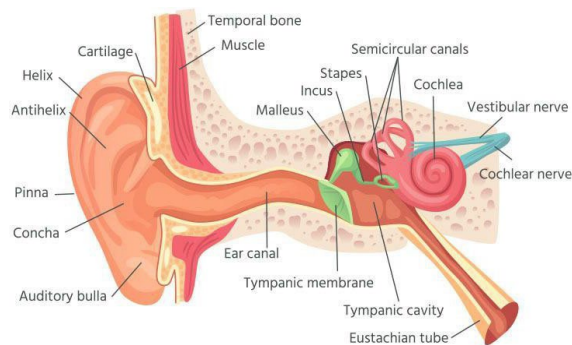


- Lasik Surgery – used to restore vision defects
 - Individual eye mapping -> ut cornea via lade -> laser used to change curvature of lens to correct refractive index
- Laser Eye for cataracts
 - Involves using a laser yo create a small cut on the edge of the cornea and inserting a ultrasonic probe into the lens capsule, where ultrasound is used to decompose proteins in the lens

Evaluation

- Allows individuals to be independent
- Improves quality of life by being able to restore sight

THE EARS



Structures and their Functions

Structure	Anatomy	Function
Pinna	Fleshy part of outer ear	Collects and directs sound waves into auditory canal, like a funnel Also shaped to help detect source of sound
Auditory Canal	Slightly curved tunnel made of bone/skin	Direct SW directly to tympanic membrane
Tympanic membrane	Very thin and transparent membrane between outer ear and middle ear	Vibrates at same freq. as the sound then passes this onto the ossicles
Ear ossicles	3 small bones in the middle ear	Amplify pressure of SW as they further transmit then to oval window of cochlea
Oval window	Thin flexible membrane of cochlea	Transmits vibrations from the stirrup ossicle to the fluid filled cochlea
Cochlea	Snail shaped, spiral tube	Inside cochlea, vibrations in the fluid will cause the hair cells (organ of corti) to move
Organ of Corti	Membrane and fluid rows of hair cells	Hair cells containing receptor "stereo cells" Push against the tectorial plate Hair cells release nerve transmission and convert mechanical energy to electrochemical

Auditory nerve	Collection of neurons that connect cochlea to brain	Transmit electrochemical message to brain for interpretation of sound
Eustachian Tube	Tube that connects middle ear to pharynx	Equalises air pressure of both sides of ear drum

Sound wave in air → soundwave funnelled by pinna into auditory canal → tympanic membrane converts SW into mechanical → ear ossicles vibrate in middle ear → oval window sets up pressure waves in cochlea → hair cells in organ of corti detect SW → neurons transfer nerve impulse to brain → brain interprets message as sound

Hearing Disorder

- Conductive hearing loss:
 - Hearing loss associated with SW not being able to reach the inner ear (ie. lack of conduction)
 - Caused by earwax buildup, benign tumour, middle ear infections, perforated ear drum
- Sensorineural hearing loss:
 - Defects/damage to the cochlea (ie. hair cells), often permanent
 - Caused by viral infection, side effects from drugs, physical trauma to cochlea, mutated gene, loud constant exposure to noise, age

Technologies

- Hearing Aids
 - Battery-powered electrical devices that are used to amplify sound in the ambient environment, such that the affected individual, who is hearing-impaired, can hear better
 - Attached onto the external ear so that it can amplify the frequencies at which the affected individual has difficulty hearing in
 - Helps with mild to moderate hearing loss
- Bone Conduction Implants
 - Works by transmitting sound vibrations through the bone
 - Amplify a sound via bone conduction, or vibrations through the bones of the skull, which directly stimulate a functioning cochlea
 - Bypass the structures of the external and middle ear and directly stimulates the cochlea in the inner ear
 - Helps if eardrum or cochlea are damaged
- Cochlear Implants
 - A small, complex electronic device that can help to provide a sense of sound to a person who is profoundly deaf or severely hard of hearing
 - Consists of an external portion that sits behind the ear and second portion that is surgically placed under the skin

Evaluation

- Hearing Aids

Benefits	Limitations
Able to restore hearing to affected individual with cochlear hair cell receptors having abnormally low sensitivity	Does not heal hearing-impaired individuals
Can aid with loss of hearing due to ear infection and head trauma	If cochlear hair cell receptors are completely damaged or dead, no amount of amplification from the hearing aid will help
Hearing aids linked to reduced cognitive decline	May amplify all sounds, including background noise, that you may not want to hear
	Usually entail an adjustment period that may take several months

- Bone Conduction Implants

Benefits	Limitations
Better hearing, even in noisy situations	May remove too much 'junk noises' eg. in dangerous events
Improved sound quality	With many procedures comes risk of infection and inflammation
Improved speech recognition (compared to hearing aids)	Small risk that the growth of new bone around the implant will fail

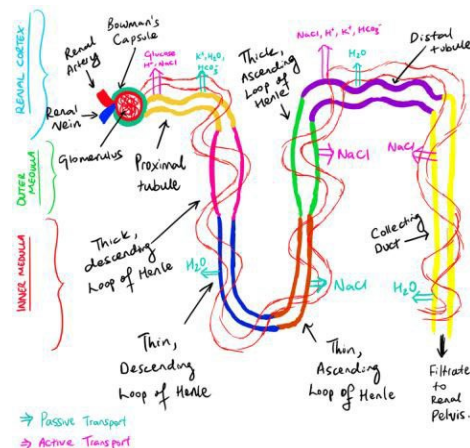
- Cochlear Implants

Benefits	Limitations
Ability to hear their own voice/improve their voice tone and pronunciation	Cost of the cochlear implant is high – ranging from \$30k to \$50k in the absence of insurance
Ability to hear sounds around them for safety and better quality of life	The actual sound frequencies will be different to that in the environment which means there is often long period of time of learning, therapy and adjusting to sounds after the surgical implantation. Sometimes can take years Results vary
Easier communication	Loss of what's left of the natural hearing

THE KIDNEYS

Structures and their Functions

- Function of kidney [osmoregulation]:
 - Controls excretion of nitrogenous waste [from breakdown of proteins and a.a]
 - Waste removal and regulation of water/electrolytes



Name	Description
Glomerulus	<u>Filtration</u> Blood cells and large proteins cannot pass through glomerulus Water and solutes passively pass through glomerulus as a filtrate
Bowman's capsule	Contains filtrate [filtered substances out of blood]
Proximal convoluted tubule	<u>Secretion</u> Active transport of unwanted substances (eg. drugs) from blood into tubule <u>Reabsorption</u> Primarily passive return of water, ions and glucose from filtrate to blood
Descending loop of Henle	<u>Reabsorption</u> Main site of water reabsorption back into blood
Ascending loop of Henle	Permeable to ions
Distal convoluted tubule	<u>Reabsorption</u> Further reabsorption of ions is regulated by antidiuretic hormone (ADH)
Collecting duct	Collecting of fluid with waste Connected to ureter to be stored in bladder

Process	Substances	Direction of Movement	Passive or Active?	Desc. Of Process
Filtration	Everything in the blood that is smaller than a RBC	Glomerulus blood vessel →nephron	P	Movement of blood in close proximity to the permeable Bowman's capsule Means substances smaller than RBC move into filtrate
Reabsorption	H ₂ O, ions, glucose, a.a, peptide	Nephron → blood	P (H ₂ O, Cl ions, etc.) A (glucose, a.a, peptides)	Moves nutrients and H ₂ O back into bloodstream
Secretion	Drugs, toxins, waste products, ions, excess water	Blood →Nephron	A [as it is selective] Some P due to diffusion and osmosis	A transport to remove any further waste into nephron prior to excretion
Excretion	Physical process of forming urine			
Hormone	Released by	Action on nephron	Effect on blood	Effect on urine
ADH	Pituitary glands	Increased H ₂ O retained	More dilute = decrease conc. ions	Increased conc. of solutes
Aldosterone	Adrenal glands	Increased reabsorption of sodium ions in blood	Increased conc. Na ions	Increased conc. of water

Technologies - Kidney Dialysis

Dialysis Overview

- Type of treatment that helps your body remove extra fluid and waste products from your blood when your kidneys are not able to
- First successful in 1940s and became standard treatment for kidney failure starting in the 1970s
- Can be done in a hospital, a dialysis centre or at home

Uses

- Acute kidney injury (AKI)
 - Sudden episode of kidney failure or kidney damage that happens with a few hours or days
 - Usually treated in a hospital with IV
- Kidney Failure (ESKD – end stage kidney damage)
 - When 10-15% of your kidney function remains
 - At this stage, kidneys are no longer able to keep you alive without extra help
 - Dialysis is only able to do some of the work of healthy kidneys, not a cure
 - Will need dialysis for the rest of your life or until kidney transplant

How it works

- Performs some of the duties that your kidney usually does to keep your body in balance
 - Removing waste/extra fluids in your body to prevent them from building up in the body
 - Keeping safe levels of minerals in your blood, such as potassium, sodium, calcium and bicarbonate
 - Helping to regulate your blood pressure

Evaluation – Kidney Dialysis

Benefits	Limitations
Treatment can be completed in comfort of home	Not a cure – treatment is ongoing and without, the patient dies
Ensures maintained function of kidney if kidneys fail	Requires many hours of treatment so significantly impacts patients lifestyle and ability to work
Highly effective in driving osmoregulation to a safe, normal level	Patient is required to undergo dialysis on a regular basis

All in all, renal dialysis is **effective in** performing its role of acting as an ‘artificial kidney’ for patients suffering from dysfunctional kidneys. This allows the patient to perform everyday functions without experiencing many symptoms that the absence of renal dialysis would otherwise render performing certain daily activities difficult. This include breathing normally. So, the advantages outweighs the disadvantages.